Playlist-Induced Spillovers: Examining On- and Off-Platform Outcomes for Emerging Artists on Spotify

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Abstract

Digitization and streaming platforms-based activities can create opportunities for artists at all stages of their careers. This paper studies the extended multiplier effects of editorial playlisting on both on-platform and off-platform outcomes for emerging, independent artists. Utilizing a unique dataset from Spotify's Playlist Pitching tool, a feature that allows artists to pitch their tracks to music editors for playlist consideration, we employ a matched difference-in-differences methodology to compare outcomes between artists who were featured on playlists and those who were not. Our results show that emerging independent artists featured on Spotify's editorial playlists not only experience a significant increase in stream counts but also have a greater likelihood of going on tour. This research emphasizes the role of digital platforms as potent discovery mechanisms that enable fan engagement, in turn, driving off-platform career trajectories including live event opportunities.

1 Introduction

Digitization and streaming platforms have increased publication and audience-building opportunities for music artists, enabling the addition of a long tail of musical content, greater entry by musicians, and new discoveries for consumers (Aguiar et al., 2023). Lower costs from digitization, through lower costs of producing, promoting and distributing new music, lead to substantial consumer welfare gains (Aguiar and Waldfogel, 2018). Some of these gains come from discovering unexpected "hits" within the larger pool of music, which now includes new entrants. Streaming platforms enable this discovery through a diverse set of mechanisms (e.g., search, playlists, promotions, recommendations) that can connect emerging artists and potential listeners who might like their music (Brynjolfsson et al., 2006). Whether connections such as playlisting lead to artist fanship is less clear, with industry executives voicing concerns that they promote passive listening (Morgan, 2020).

While digitization has dramatically increased the production and distribution of recorded music by entrants, its impact on the live performances market is less well understood, and this matters for musicians. Despite streaming serving as the main revenue source for the music industry since 2017 (IFPI, 2023), live performances remain an important source of earnings for emerging artists musicians (DiCola, 2013; Krueger and Zhen, 2018; Webster et al., 2018; MIDiA Research, 2019; Hesmondhalgh et al., 2021; MIDiA Research, 2024), and may also serve to deepen fanship. Live performances are typically managed by concert promoters, with booking agents acting as intermediaries between promoters and artists. Typically, emerging artists lack booking agents and promoters will not book directly with artists (Apruzzese et al., 2023).

Spotify's editorial playlists, curated by human experts, serve as a nice empirical setting to study one of the many ways in which streaming platforms enable the discovery of emerging artists and how this discovery affects their career trajectory on- and off-platform. Using unique proprietary data, we employ a difference-in-differences framework to compare matched emerging artists with new releases before and after submitting their music for consideration to an editorial playlist. For our period of inquiry, October 2021 and September 2022, we find that playlisting causes a dramatic increase in streams and on-platform engagement, such as follows, shares, and saves to a user's playlist, as well as a notable increase in the likelihood of going on tour. Our results suggest that editorial playlists serve as an impactful mechanism for discovery, having a multiplier effect on artists' on- and off-platform development of an enduring fanbase.

One plausible explanation for our findings is that the studied editorial playlists can magnify the reach of new releases, renew engagement with established fans, and expose an artist's music to a broader audience. Externally, this enhanced recognition can make artists more commercially appealing and serve as a signal of credibility to persuade venues, promoters, and booking agents of their viability as a live act, increasing the prospects of live event bookings. Our work joins a stream of important literature (Mortimer et al., 2012; Papies and van Heerde, 2017; Christensen, 2022) that examines the complementary relationship between online and live music consumption. We bring an important, exclusive focus on music entrants and how digitization affects their ability to build a music career.

1.1 Discovery, Engagement, and Performance in the Digital Music Economy

Our work is most closely related to the literature on product discovery in digital markets. Digitization has driven an increase in the supply of available content, which has proven to be highly beneficial to consumers (Brynjolfsson et al., 2003; Aguiar and Waldfogel, 2018). The proliferation of content has led online platforms to serve not only as distributors, but also as major product recommendation and discovery mechanisms. One important mechanism for discovery are platform managed lists, which can take the form of bestseller lists —driven by demand— or algorithmic and editorial curated lists (Aguiar et al., 2023).

Recent studies have explored the effect of platform managed lists on the consumption of digital media. Closely related to this study, Aguiar and Waldfogel (2021) measure the effect of tracks

being added to Spotify's most prominent editorial playlists, and find a significant multiplier effect on the economic success of the songs. Similarly, Sim et al. (2022b) use a regression discontinuity design to analyze the multiplier effect of entering the top 100 charts of a major music streaming service in South Korea, and find an increase of 11-13% new listeners along with a 2-3% increase in streaming counts, which they attribute to a saliency effect rather than observational learning.

A growing body of research has also explored the complementary relationship between the demand for recorded music and live music, and how it has evolved with the emergence of digital platforms. Christensen (2022), for example, exploited a natural experiment involving the temporary removal of Warner Music's catalog from YouTube in 2009. The study demonstrated a significant negative impact on the relative revenue, ticket sales, and pricing of live concerts for Warner artists compared to non-Warner artists, offering robust causal evidence that streaming platforms boost demand for live performances. Similarly, Papies and van Heerde (2017) examined the relationship between record sales and concert revenue for 387 of Germany's most successful artists, finding that a positive feedback loop exists between the two: strong performance in one area bolsters the other. However, their results revealed an asymmetry: record sales had a stronger influence on concert revenue than vice versa. These studies focus primarily on well-known, established artists, motivating further analysis of digital platforms' impact on emerging musicians.

Other research has considered lesser-known artists. Mortimer et al. (2012) examined the effects of file-sharing platforms introduced in 2000, reporting declines in album sales but an increase in both the demand for and supply of live performances, especially for emerging and lesser-known artists. Cho et al. (2017) provided corroborating evidence, documenting a rise in the number of concerts and their geographic dispersion from 2000 to 2011.

A set of studies considers how listener behavior changes after attending a live performance, suggesting a deepening of listener fanship. They find a complementary relationship between live concert attendance and subsequent streaming and social media engagement (Danielsen and Kjus, 2019; Montoro-Pons et al., 2023; Hogue, 2024). Unlike these studies, which focus primarily on post-concert activity, our research examines the reverse dynamic: how listener discovery via editorial playlisting affects the likelihood of emerging artists to give a live concert, creating a multiplier effect.

Our work extends this literature two key ways. First, most work focuses on the effects of best selling charts and major editorial playlists, which usually feature well-established artists. We are intentional in examining the effects on creators in the long tail of the streaming distribution, entrants made possible from digitization. Second, we go beyond streams to assess a wide range of outcomes within the platform that relate to user discovery and fan engagement. Most notably, we are able to observe a measure of touring, which allows us to examine how playlisting affects off-platform outcomes for artists, as well as saves, follows, and user playlisting, which suggest deepening fanship.

2 Institutional Context: Spotify Playlist Ecosystem

In the following section, we provide a background on the Spotify Playlists Ecosystem, the Playlist Pitching mechanism, and the role of live shows in artist revenue.

2.1 Spotify Playlists

Playlists on Spotify can be classified into four main categories, depending on who creates them: editorial playlists, algorithmic playlists, artist and listener playlists. This section describes each type of playlist, beginning with editorial playlists, which are the focus of this paper because they are the best suited for empirically estimating spillovers from being playlisted on Spotify, e.g. they have properties that help address confounding.

Editorial playlists are lists curated by Spotify's editorial team. Editorial teams are usually made up of curators with extensive experience in the music industry, including former A&R executives, music journalists, radio programmers, and professional musicians (Shah, 2017). These playlists are usually based on a genre, can be mood-and activity-based, and serve specific segments of creators on the platform. Spotify's playlist "ecosystem" is composed of thousands of editorial playlists from all around the world, with a varying number of "followers," ranging from more than 30 million to a few hundred. Among the most popular playlists (i.e. the most followed) are Today's Top Hits, RapCaviar, and Viva Latino. Of particular interest for this study are playlists that have a special focus on emerging artists, such as the Fresh Find series, which features new music exclusively from independent artists and labels.

Algorithmic playlists are created by recommendation algorithms that, according to Spotify, "rely on a number of inputs. The importance of these inputs may vary over time, depending on your individual use of Spotify... As you engage with Spotify, actions such as searching, listening, skipping, or saving to Your Library influence our interpretation of your taste... Recommendations are also based on information you share with Spotify, like your general (non-precise) location, your language, your age, and who you follow" as well as trends, information about the content (such as genre, release date, etc.), and listener safety (such as potentially violating content) (Spotify, 2024). Some of the most popular algorithmic playlists are Discover Weekly—a playlist that serves users 30 tracks they have not played before and refreshes every Monday—and Release Radar, which contains the latest releases from artists followed by the user as well as recommended artists, and refreshes weekly.

Lastly, artist and listener playlists are created and managed by artists on Spotify and Spotify users. Artist playlists can be pinned to their artist page in Spotify for Artists. Listener playlists vary greatly, and although there are some that can amass millions of users, they are created mainly for personal music consumption. These playlists, as well as algorithmic playlists, are not the focus of this paper.

2.2 Spotify's Playlist Pitching Tool

Spotify's Playlist Pitching tool for editorial playlists was launched on July 19, 2018, with the objective of giving artists, labels and management teams a streamlined process for pitching a song for playlist consideration (Spotify, 2018). The motivation behind the mechanism is to provide artists the opportunity to highlight their music to the appropriate editor for review, by submitting relevant metadata from their new tracks. Spotify argues that this will increase the likelihood that the new music is showcased to the appropriate playlist, and consequently to the appropriate audience. During its initial beta phase, which lasted until October 24, 2018 (3 months) around 67k submissions were made, from which close to 10k were included in an editorial playlist (Perez, 2018).

During our period of inquiry, between 2021-2022, the pitching process unfolded as follows: artists submit a single new track, filling out a form detailing both the musical characteristics and the cultural context of the track and the artist. The editors review these submissions and make a decision on whether to include the track in an editorial playlist. Once a song is live on the platform, it is no longer eligible for pitching.

2.3 The Role of Live Shows in Artist Revenue

Several surveys conducted between 2012 and 2024 found that live shows are an important source of income for both career-aspiring musicians, a description that fits the emerging artists in our sample, as well as professional musicians. The average revenue earned from touring estimates vary between 28-42 percent depending on the surveyed population (DiCola, 2013; Krueger and Zhen, 2018; Webster et al., 2018; MIDiA Research, 2019; Hesmondhalgh et al., 2021; MIDiA Research, 2024), and the degree to which respondents consider themselves professional musicians, with professional musicians earning a greater income share than career-aspiring musicians from live shows. For booking shows, there are broadly two main parties involved, the artist's management team and the concert promoters. An artist manager or tour agent would usually have the task of selling and negotiating shows with concert promoters. The concert promoter then has the task of organizing the show by booking the musician, securing an adequate venue for the show, contracting with the ticketing company to sell and distribute tickets, and advertising the event (Krueger, 2019). Concert promotion in the US, especially for the bigger tours, is mostly consolidated in two main players: Live Nation and the Anschutz Entertainment Group. Independent, emerging artists typically do not have a dedicated booking agent, unlike signed and/or established artists (Apruzzese et al., 2023).

3 Data

The data for our analysis consists of internal proprietary data from Spotify. The summary statistics presented in this section correspond to a pool of matched artists restricted to being independent,

emerging, and not previously featured in an editorial playlist. We justify these restrictions in detail in Section 5.2 .

We compile a dataset of all new music submissions through the Playlist Pitching tool from October 2021 to September 2022. Our dataset consists of track-level submission forms, detailing both the musical characteristics and metadata of the track and the submitted information about the artist. The dataset encompasses submissions from artists in all markets where Spotify operates. In addition, we collect the audio features of the submitted tracks. These features include 13 high-level descriptors provided by Spotify, and musical elements such as the track's key, tempo, and instrumentalness.

For the same period, we also observe all additions to Spotify's editorial playlists. In particular, we observe the playlist's name, the date of addition, and the position of the track on the playlist. All of the recorded additions are the result of an editorial decision by one (or more) of the hundreds of editors into thousands of editorial playlists on Spotify. We also observe metadata related to the playlist, such as its monthly active users, the principal markets they serve, their main genre or topic, and if they are part of a series (such as New Music Friday).

For the set of artists with submissions in the period of study, we observe time series data of their monthly total streams and the royalty generated by their music on Spotify. We provide summary statistics of monthly streams and corresponding royalty data in Table 1 Panel A, at the artist-month level. In addition to streams, we observe the artist's monthly listeners, followers, and track engagement metrics such as user-generated playlist adds and shares to social media sites. The artist's monthly listeners is the number of unique Spotify users that streamed the artist for at least 30 seconds. Similarly, monthly streams record the total number of streams that the artist's tracks received in a given month. The followers count corresponds to the total number of users that "follow" the artist in Spotify, which is achieved by clicking on the follow button in the artist's page. The number of shares corresponds to the total number of times the artist's tracks were intended to be shared to social media, measured by users clicking on the share option of a track. Finally, we observe the number of additions to user-generated playlists. We provide a summary of these metrics at the artist-month level in Table 1 Panel C.

We are able to measure live show activity in two different ways: through Spotify's internal tracking data of artist touring activity and through the artists' Spotify concert page. For the latter, we are able to observe a binary indicator that is equal to one if an artist has a show in the last 30 days or anytime in the future. Throughout the paper, we refer to this outcome as "On Tour." We also track activity within the artist's concert page. This page showcases artists' upcoming shows, and is only visible to users when there is at least one show in the future registered on the platform. Although we are not able to observe the show information, we can track user visits to the artists' concert page and click-through activity within the page. In particular, we observe the action of clicking through the "Find tickets" button that links to an off-platform live show ticketing partner. We provide summary statistics of these data in Table 1 Panel B.

For artists in our sample, we are able to observe their metadata and relevant information

Table 1: Summary Statistics: Artist–Month Level Outcomes.

	Mean	Std. Dev	Median	P5	P95
Panel A: Streams & Royalties					
All Streams	14,095	146,358	1,293	25	56,558
All Streams (Log)	7.16	2.33	7.17	3.27	11
Total Royalties USD	47	490	3.87	0.06	193
Total Tracks with Streams	19	32	10	2.00	60
Panel B: Live Show Metrics					
On Tour	0.05	0.23	0	0	1
Total Concert Page Views	16	1,091	0	0	8.00
Total "Find Tickets" Clicks	1.64	31	0	0	0
Has Concert Page Views	0.06	0.24	0	0	1
Has "Find Tickets" Clicks	0.04	0.21	0	0	0
Panel C: On-Platform Engagement Metrics					
Followers	640	2,163	133	10	2,656
Monthly Listeners	6,471	56,560	470	10	25,314
Track Shares	43	883	6.00	0	125
User-generated Playlist Adds	380	5,399	34	0	1,322
Followers (Log)	4.99	1.65	4.90	2.40	7.88
Monthly Listeners (Log)	6.23	2.42	6.15	2.40	10
Track Shares (Log)	2.03	1.58	1.95	0	4.84
UG Playlist Adds (Log)	3.62	2.14	3.56	0	7.19
Has Track Shares	0.81	0.39	1	0	1
Has User-generated Playlist Adds	0.93	0.26	1	0	1

Note: artist–month level statistics computed for our sample of 12,150 artists observed over 14 months.

regarding their release activity. In particular, we observe the artist's main genre and the year of their first release. Regarding their release activity, we observe, at a given point in time, the artist's latest release date; whether the artist has worked with a major licensor as opposed to an independent label or self-release; and the artist's rank on the platform, based on streamshare.

Using the data described above, we construct the dataset for our main analysis. We create a panel dataset, where the unit of observations is at the artist-month level. This way, for each artist, we track their on-platform activity (number of streams, followers, unique listeners, shares, playlist adds, and concert page activity) as well as their off-platform activity (touring). These outcomes are described in Table 2. Uniquely, we are able to identify the dates when artists submit a new song through the Playlist Pitching tool for playlist consideration.

We will use the month when the new track is submitted to the Playlist Pitching tool as our treatment month. We make this choice because it allows us to align both the treatment and the control group in relative time in our empirical strategy. Alternative definitions, such as the time of addition to the playlist, do not work as well with our empirical strategy, given that the tracks in the control group do not have a comparable date. Furthermore, on average, songs are added to playlists 18 days after being submitted, and 86 percent of playlisted songs are added within 30 days

of being submitted. In the appendix Table A7 we consider the track release date as an alternative treatment month, which yielded similar results.

Given that we are interested in observing how the outcomes change after a new release submission and potential inclusion in an editorial playlist, we construct a time series including data for 7 months before and 7 months after submission.

Table 2: Description of Outcome Variables of Interest.

Outcome	Description	Range
All Streams (Log)	Total streams in Spotify for all artist's tracks	2021-03 to 2023-03
Old Music Streams (Log)	Total streams in Spotify for all artist's tracks released up until 5 months before the new submission	2021-03 to 2023-03
Followers (Log)	Total number of followers	2021-03 to 2023-03
Monthly Listeners (Log)	Number of unique listeners streaming at least 30s of an artist's track	2021-03 to 2023-03
Track Shares (Log)	Number of clicks in the "share" button for any of the artist's tracks	2021-04 to 2023-03
User Playlist Adds (Log)	Number of times an artist's tracks were added to user-generated playlists.	2021-03 to 2023-03
On Tour	Binary indicator equal to 1 if an artist has a live show in the last 30 days or anytime in the future	2021-03 to 2023-03
Has Concert Page Views	Binary indicator equal to 1 if an artist has at least 1 view to their concert page	2021-03 to 2023-03
Has "Find Tickets" Clicks	Binary indicator equal to 1 if an artist has at least 1 click on the "Find Tickets" button for any of their upcoming shows	2021-07 to 2023-03

Note: All outcomes are at an artist-month level.

4 Mechanism

In this section, we discuss the hypothesized causal pathways that lead a playlisted artist to see increased touring activity and increased fan engagement. As discussed earlier, playlists in the Spotify ecosystem are tailored to various objectives based on their reach and type. Notably, a subset of these playlists, like "Fresh Finds," is dedicated to spotlighting emerging artists, specifically those who are not signed by a major label. Such playlists can be pivotal for artist and musical discovery and are the main interest of this study.

When an artist's track is added to such a playlist, the immediate consequence is a heightened exposure to Spotify's listeners, driven by the playlist's followers, as evidenced in Aguiar and Waldfogel (2021). Given the editorial choices where tracks are matched with playlists based on elements like genre, culture, and language, featured tracks are likely to be more visible to a potentially receptive audience. Consequently, being featured on a playlist is hypothesized to not only increase the track's streaming counts in the short term but also serve as a discovery mechanism. In particular, the playlist will showcase the artist's music to both new and old listeners. While it renews engagement with existing fans, it simultaneously acts as a discovery platform for potential new

fans.

The most direct impact of a track's playlisting will be in the streaming counts for the playlisted track. We expect to see a direct increase in streams as users listen to the playlist while the track is featured. If or once the track is removed from the playlist, a sustained increase in streams might happen if listeners developed an interest in the track and continue to stream it. Moreover, playlist additions may drive Spotify users to explore the artist's catalog beyond the playlisted track. Therefore, we predict an uptick in streaming activity for the artist's other songs, encompassing both earlier releases and newer tracks released concurrently with or after the featured song.

The process of discovery resulting from playlisting may lead to increased user engagement with the artists on the platform. This engagement can be reflected through different user behaviors. For instance, more users may "follow" the artist on the platform. This not only signals user interest, but also prompts Spotify to recommend more of the artist's content to the user. For example, "following" an artist will mechanically cause the user's algorithmic "Release Radar" playlist to feature the followed artists' new music. Artists can also experience growth in the number of their tracks added to user-curated playlists, a further indication of active discovery. Additionally, users might share the artist's music on other social platforms, further expanding the artist's reach and potentially increasing demand.

The increased demand for the artist's music and heightened platform engagement can influence both the supply and demand side of the artist's live performances. Artists are more often willing to play live than they are able, so there is excess supply of artists willing to perform a live show. Addition to a playlist — followed by the consequent rise in streams and engagement metrics — can elevate the artist's recognition, making them more appealing to concert promoters and venues. It also serves as a signal of credibility to convince stakeholders of the artist's viability as a live act, which may lead to an increased frequency of bookings post-playlisting. On the demand side, the enhanced visibility could drive Spotify users and fans to the artist's live shows.

5 Empirical Strategy

We take a first look at the data to illustrate the effect of the playlist in the track's streams. Figure 1 shows the daily streams of a track featured in the editorial playlist "Fresh Finds Spain" over the first 40 days following its release on August, 2022. Key events are marked: the track's addition to the playlist five days post-release is indicated by a black dotted line, and its removal three weeks later by a red dotted line. The graph shows a sharp discontinuity in the number of streams at these points, with an increase in streams the day after the track's playlist inclusion and a decrease following its removal.

To more rigorously assess the effect of playlist inclusion, we analyze a dataset comprising artists whose songs were either included in playlists or not. However, direct comparisons between playlisted and non-playlisted artists may introduce bias due to likely nonrandom selection into playlists. It is plausible that playlist curators choose artists who exhibit upward trends, anticipating that these

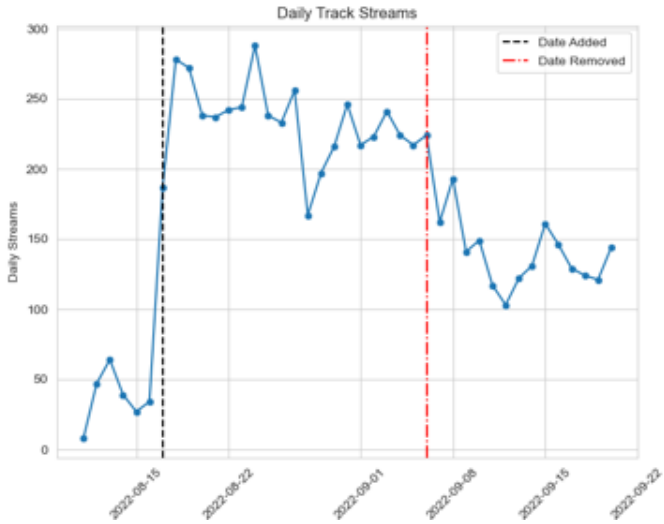


Figure 1: Daily Streams of Song Added to Fresh Finds Spain.

Note: The figure shows the total number of daily streams for a song that was playlisted in Fresh Finds Spain for 21 days. Key events are marked: the track's addition to the playlist five days post-release is indicated by a black dotted line, and its removal by a red dotted line.

artists can enhance engagement both with the playlist and, consequently, the platform. As a result, these artists might have better streaming and concert performance, regardless of whether they are being playlisted. We use two empirical strategies to control for such endogeneity: we pre-process our data by restricting our sample to emerging and independent artists and then use matching to identify an adequate control group. We then employ a difference-in-differences design to estimate the effect of playlisting.

5.1 Difference-in-Differences

We use a stacked difference-in-differences specification (Cengiz et al., 2019; Baker et al., 2022; Roth et al., 2023) to estimate the effect of playlisting on both on- and off-platform activity:

$$Outcome_{itg} = \delta(\text{Playlisted}_{ig} \times \text{After Submission}_{tg}) + \alpha_{ig} + \lambda_{tg} + \epsilon_{itg} \tag{1}$$

Where $Outcome_{itg}$ indicates any one of the artist-month-level outcomes described in Table 2. For the non binary outcomes, we transform the dependent variable by taking the log of (Y+1). In the expression, i indexes the artist, t indexes the year-month, and g indexes submission-month cohorts in a stacked-regression design. Playlisted_{ig} is a binary indicator equal to one if artist i was playlisted in the stacked dataset. $\text{After Submission}_{tg}$ is an indicator that is set to one for periods after the submission of a new song. We include α_{ig} and λ_{tg}, which correspond to artist and year-month fixed effects with index g for the specific stacked dataset. We do not include time-invariant

artist covariates, given that they will be perfectly correlated with the artist fixed effect. We cluster the standard errors at the artist level. We have a balanced panel of 12,150 artists measured 7 months before and up to 7 months after the new song submission.

To examine the parallel trends assumption, we estimate the following specification for an event study:

$$\begin{aligned} Outcome_{itg} = & \sum_{j=-6}^{-2} \delta_j (\text{Playlisted}_{ig} \times \mathbf{1}(t - E_i = j)) \\ & + \sum_{j=0}^{6} \delta_j (\text{Playlisted}_{ig} \times \mathbf{1}(t - E_i = j)) \\ & + \alpha_{ig} + \lambda_{tg} + \epsilon_{itg} \end{aligned} \tag{2}$$

Here, E_i indicates the year-month of submission for artist i. $\mathbf{1}(t - E_i = j))$ is an indicator for being j months from the submission month. We omit the pre-periods corresponding to -1 and -7 months before submission in our estimation (Borusyak et al., 2023).

5.2 Subset and Matching

To address endogeneity concerns, we narrow the set of artists that we consider for the analysis. Our emphasis is on lesser-known, independent, and emerging artists who have a new release and have never been playlisted. We focus on lesser-known independent artists because artists affiliated with major licensors benefit from superior production, promotion, and marketing resources as compared to those on an independent label or who are self-produced. Better production and promotion may lead to better streaming and concert performance after a new release, regardless of whether they are playlisted. Moreover, lesser-known artists are also more prone to adjust their touring schedules within a shorter time frame, unlike their more established counterparts, who often schedule shows well in advance. Narrowing our sample in this way limits confounding from preexisting differences in artists and their releases. Our restriction to new releases is in response to work showing that new releases increase sales of older music (Hendricks and Sorensen, 2009), and ensures that all artists are at the same point in the release cycle. Finally, we focus on artists who have never been playlisted before because being playlisted likely increases streams, exposure, and live shows. This means the response to a new release for a previously playlisted artist may be larger than for a non-previously playlisted artist, regardless of whether they are playlisted for this release.

Spotify editors often prioritize newly released or unreleased tracks for curated playlists. Thus, Spotify's Playlist Pitching tool provides a great opportunity to study artists who recently debuted their music and seek to be added to an editorial playlist. This mechanism allows artists with new content to propose their latest tracks for editorial consideration. We analyze the data for 226,526 emerging artists (where 7,746 are playlisted) that fit the description above, and submitted new music to Spotify and the Playlist Pitching tool in the period from October 2021 to September

2022, to circumvent the decline in live shows during the COVID-19 pandemic, as suggested in Figure A1. A desirable consequence of conditioning on new release and submission is that all artists in our sample share the intent to be added to an editorial playlist. It is a separate decision-maker (the editorial team) that decides the treatment status with the information available, specifically the submission form, the artist's previous performance on Spotify, and their own idiosyncratic taste.

The resulting set of playlisted artists is featured mainly in the "New Music Friday" and "Fresh Finds" playlists (56%), which are playlists that are mostly focused on emerging artists and new music. We provide a breakout summary of the playlist by types defined by Spotify in Table A1.

This narrowing process gives us a more comparable set of artists. Plotting the outcomes for the periods before the new submission reveals almost identical trends between playlisted and non-playlisted artists in our outcomes of interest and other observable characteristics. However, the trends and summary statistics reveal a difference in levels for almost every outcome, which might be concerning for our estimation (Kahn-Lang and Lang, 2020). We thus further control for confounders by using matching to pre-process our data (Ho et al., 2007). In particular, we identified a set of non-playlisted artists that are similar in observed characteristics to the playlisted artists, drop the unmatched artists in the control group, and then use OLS to estimate the parameters in Equation 1.

We use propensity scores for each artist, that is, the probability of being playlisted as estimated with a logistic regression, to match playlisted and non-playlisted artists. We implement propensity score matching as follows, first estimating the scores for each artist using a logit model:

$$Pr(\text{Playlisted}_i = 1|X_i) = \frac{e^{\beta \cdot \mathbf{X}}}{1 + e^{\beta \cdot \mathbf{X}}} \tag{3}$$

Where Playlisted is a binary variable that indicates whether the artist was playlisted during the period of study, $\mathbf{X}$ is a vector of time-invariant observable characteristics related to the artist and the new track submission form, as well as lag outcomes for the periods -6 to -1 before submitting a new release, and β is a vector of regression coefficients. Given the potential ability of the editors to pick up on unobserved musical quality, we also include 13 measurements describing the musical structure of the track, computed by Spotify, which include measures of mode, key, tempo, time signature, among others. We do exact matching on the track's submission month, as well as the track's genre, the track's language, the type of label and the artist quantile group, defined by their streamshare. We include a list of the characteristics used to compute the propensity score in the Appendix Table A2. Next we create a control group composed of non-playlisted artists, by matching them to playlisted artists using the propensity score. To identify the best control group, we use nearest-neighbors 1-to-1 matching without replacement (Austin, 2010), and we drop all the unmatched artists from the control group. We use the resulting set composed of 6,075 playlisted artists, and an equal number of non-playlisted artists to estimate the parameters in Equation 1 for all of our outcomes of interest.

6 Results

We first discuss the results for the propensity score used for matching playlisted and non-playlisted artists. Then we turn to our difference in difference specification 1, where we estimate the effect of playlisting on both on- and off-platform outcomes. Finally, we explore heterogeneity in the effects of playlisting across artist subgroups, based on genre, language, and popularity.

6.1 Propensity Score Matching

The results from the logit regression from Equation 3 confirm that playlisted artists differ from our full set of non-playlisted artists, suggesting that the editor's assignment to playlists is not random even after conditioning on lesser-known independent artists. The high correlation among streaming and engagement metrics complicates interpretation, making the full model less informative. However, we identify that playlisted artists tend to have a higher number of monthly listeners, a higher number of followers, and are relatively more new. In addition, artists from certain genres seem to be more likely to be playlisted. We also identify that tracks that are not covers, remixes, or instrumental seem to be preferred.

After matching, we see an improvement in covariate balance. In Table A3 we report summary statistics to assess the balance of the covariates and the quality of the matching. We report the means of the treatment group, and the matched and unmatched control groups, as well as the standardized mean differences. This measure is the difference in means of each covariate between the treatment and control groups, standardized by the standard deviation of the covariates in the treatment group, so it is on the same scale for all covariates (Stuart et al., 2013). Notably, the standardize mean differences between the treatment and the matched control groups are very close to zero, as is desired (Stuart et al., 2013). In terms of the 13 measurements describing the musical structure of the tracks, in the appendix Section 9.1 we find that there is no discernible difference between the playlisted and the non-playlisted tracks.

We use this matched set, consisting of 6,075 playlisted artists and 6,075 non-playlisted artists, to estimate our difference-in-differences models, as specified in Equations 1 and 2 for our outcomes of interest.

6.2 Effect on Streams

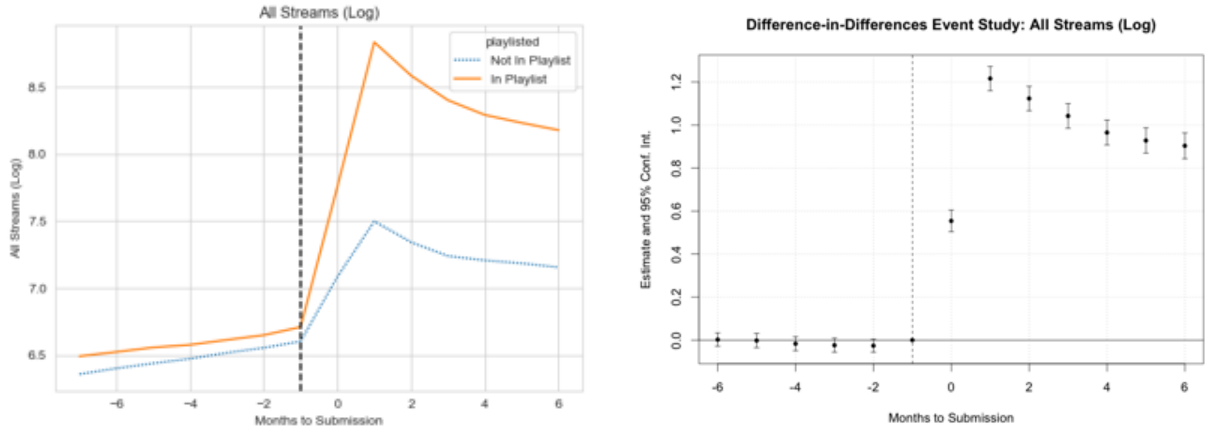
We provide model-free evidence of the effect of playlisting by plotting the average streams of playlisted artists and non-playlisted artists in Figure 3a, for 7 months before and up to 7 months after, and including month of submission of a new release to the pitching mechanism. The first thing to note is that the streams before a new release are very similar across playlisted and not playlisted artists. Then, both show an increase following a new release. This is likely due to the new song driving attention to the artist, along with the artist's promotional efforts to showcase their music, and is in line with previous literature (Hendricks and Sorensen, 2009). Secondly, we observe that the increase in streams is considerably larger for artists that are playlisted.

Table 3: OLS Regression Results for Various Dependent Variables.

	(1)	(2)	(3)
Panel A: On-platform Streams			
Dependent Variable	All Streams (Log)	Other Music Streams (Log)	Old Music Streams (Log)
Playlisted x After Submission	0.971***	0.389***	0.201***
	(0.026)	(0.022)	(0.016)
Panel B: Live Show Metrics			
Dependent Variable	On Tour	Has Concert Page Views	Has "Find Tickets" Clicks
Playlisted x After Submission	0.009**	0.010***	0.006
	(0.003)	(0.003)	(0.003)
Panel C: Engagement Metrics			
Dependent Variable	Followers (Log)	Track Shares (Log)	User Playlist Adds (Log)
Playlisted x After Submission	0.140***	0.557***	0.905***
	(0.012)	(0.018)	(0.021)
Observations (N): 170,100			

Notes: The table reports OLS estimates of the interaction between *Playlisted* and *After Submission* as specified in Equation (1). The unit of observation is at the artist-month level. Standard errors clustered at the artist level are in parentheses. All regressions include artist and month fixed effects.
Signif. Codes: *** $p < 0.001$, ** $p < 0.01$, * $p < 0.05$

Figure 2: Average Trends and Event Study for Streams.



(a) Average Log Streams of Playlisted and Non-Playlisted Artists.

(b) Difference-in-Differences Event Study Plot for Log Streams.

Note: The figure on the left shows the average log streams for artists in the treatment and control groups. The figure on the right shows the event study estimates from Equation 2. Dependent variable is the artist's monthly log streams.

We formally measure this increase with the difference-in-differences specification shown in Equation 1, where the outcome of interest is the artist's monthly log streams. We show the interaction term coefficients estimated in Equation 2 in Figure 3b and the average effects in Table 3 Panel A, column (1). In line with expectations, playlisting increases streams significantly for featured artist by 0.97 log points (equivalent to a 160% increment) relative to non-playlisted artists. We note that this increase peaks in month one after submission and then slowly decays, but persists at least over a 6 month period.

6.3 Spillovers into the Streams of Non-Focal Tracks

Even after conditioning our analysis on independent, emerging artists and pre-processing our dataset with propensity scores, there might be concerns that the observed increase in streams is driven by the superior quality of the new releases of playlisted artists, which would inherently outperform tracks by non-playlisted artists irrespective of playlist inclusion. To address this concern, we explore potential spillovers into the artists' catalog by measuring the effect on their other music, omitting the streams of the newly submitted "focal" track. Specifically, we compute two additional streaming metrics: 1) "Non-Focal" Music Streams, composed of the monthly streams of all the artist's music with the exception of the focal track, and 2) "Old" Music Streams, composed of the monthly streams for all tracks released up until 5 months before the new release submission.

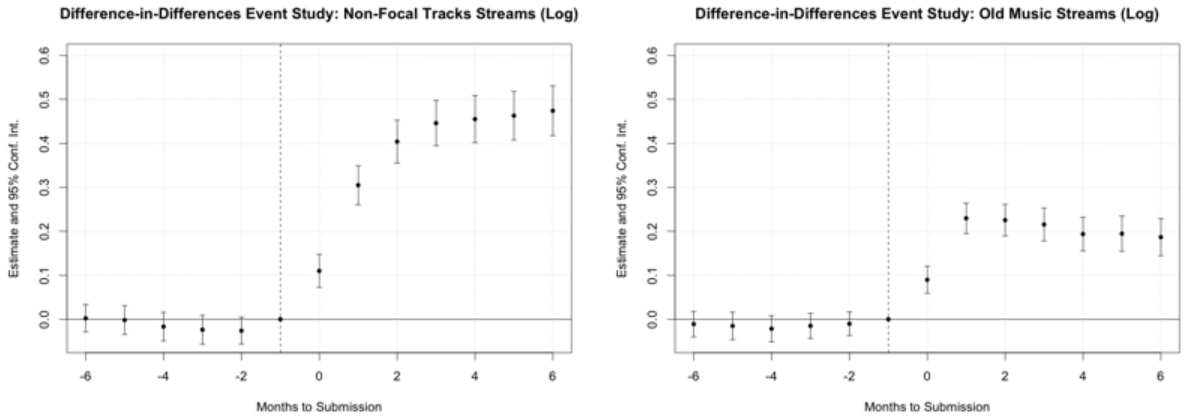
We estimate the effect on these two metrics and plot the interaction term coefficients in Figures 3a and 3b for Non-Focal and Old Music streams respectively. We also report the average effects in Table 3 Panel A, columns (2) and (3). In both cases, the increase is lower than the lift estimated including the playlisted track, but nonetheless significant. Notably, in Figure A4 we observe that for both the treatment and control groups, their "old" streams decay at similar rates up until the month of submission. Following the submission, both groups observe a short-lived rise in their streams, followed by a decay. However, the increase is significantly higher for the playlisted group and this difference persists even after 6 months post-release, as shown in Figure 3b. The findings suggest that playlist inclusion not only enhances the streams of the featured track, but also leads listeners to further explore or reengage with their prior catalog, beyond what would be expected following a new release.

6.4 Effect on Live Shows

In Table 3 Panel B, column (1), we show that, relative to non-playlisted artists, playlisted artists see an increase of around 0.9 of a percentage point in the likelihood of going on tour. The corresponding event study in Figure 4a shows a gradual increase in the likelihood, as we might expect shows to be booked and announced not immediately after a new release, as potential negotiations with booking agents and venues might not conclude until after a few weeks of an increase in fame.

On Spotify, an artist's concert page is displayed only when there is at least one upcoming show scheduled. As a result, we observe user visits to the concert page only after the artist goes on tour and the page becomes visible. Therefore, a binary metric indicating any user activity on the

Figure 3: Event Study Plots for Artists' Other Music.



(a) Non-Focal Music Streams (Log). (b) Old Music Streams (Log).

Note: The figure on the left shows the event study estimates from Equation 2 where the dependent variable is the artist's monthly streams for all songs except the Focal Song (new submitted track). The figure on the right shows the estimates where the dependent variable is the artist's monthly streams for old music, defined as tracks released up to 5 months before the new song submission.

concert page could act as a proxy for the existence of an upcoming live show, while also reflecting user interest in the artist's shows.

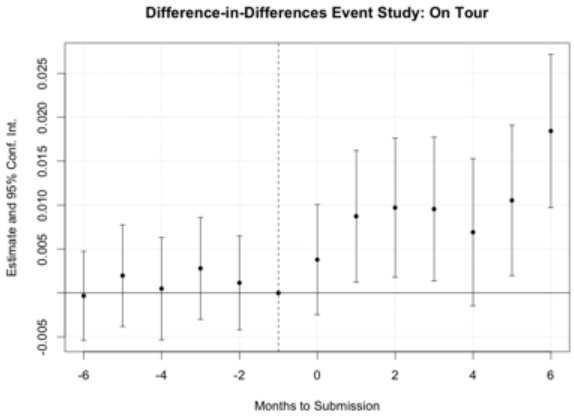
We thus examine the changes in the probability that an artist's Spotify concert page receives at least one user visit in a month. Our findings reveal that playlisted artists see an increase in this likelihood of close to 1 percentage point, as outlined in Table 3, Panel B, column (2). Notably, this uptick is close to the observed increase in the likelihood of touring. Furthermore, column (3) indicates that the chance of an artist receiving at least one click on the "Find Tickets" button for any of their shows goes up by half a percentage point. As shown in Figures 4b and 4c the coefficients show a gradual increase that closely follows that of the likelihood of going on tour.

6.5 Effect on Monthly Listeners, Followers and Track Engagement

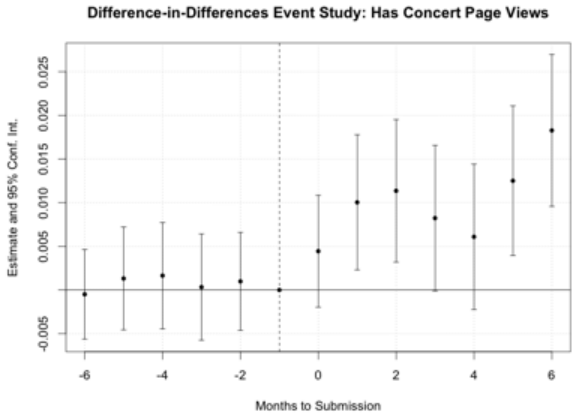
We now focus on measures of engagement within the platform that might explain the consequent increase in the likelihood of the artist's live shows. In Table 3, Panel C, we show the average effect of playlisting on monthly followers, track shares, and user-generated playlist adds. In the Appendix Figure A3, we show the event study plots for these outcomes. Playlisted artists see an average increase in followers of close to 14%, a trend that consistently rises throughout our observation period.

Alongside a significant rise in followers, tracks by playlisted artists exhibit increased engagement. As evidenced in Table 3, Panel B column (2), playlisted artists see an uptick of about 0.55 log points (or roughly 70%) in the number of times tracks are intended to be shared to other social media platforms by Spotify users. Column (3) shows there is an increase in the number of times the

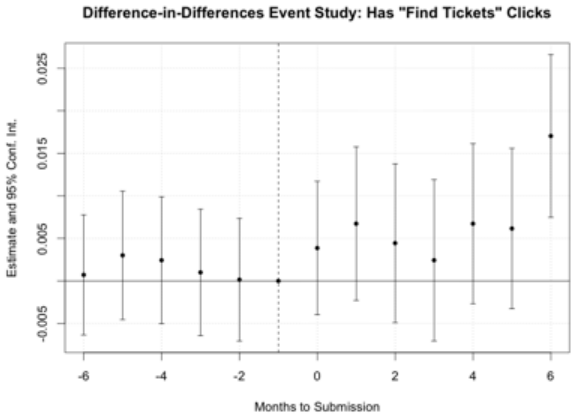
Figure 4: Event Study Plots for Artists' On- and Off-platform Live Show Metrics.



(a) Probability of Being On Tour.



(b) Probability of Having Concert Page Views.



(c) Probability of Having Find-Ticket Clicks.

Note: The figure on the top left shows the event study estimates from Equation 2 where the dependent variable is a binary indicator of whether the artist had any show scheduled in that month or any time in the future. The figure on the top right corresponds to the estimates where the dependent variable is a binary indicator of whether the artist had any user visits to their Spotify Concert Page during the month. The figure on the bottom center corresponds to the estimates where the dependent variable is a binary indicator of whether the artist had any user click on the "Find Tickets" button for concerts listed on Spotify.

artist's tracks are incorporated into user-generated playlists of 0.9 log points (or approximately 140%). Taken together, we observe that the increase in followers seems to come hand in hand with an increase in engagement with the artist's tracks, potentially driving overall artist demand and popularity.

6.6 Heterogeneity

Table 4: Estimates for Genre, Language, and Quantiles.

Category	Subset	All Streams (Log)		On Tour		N
Genre	Pop	0.939***	(0.079)	0.017*	(0.007)	17220
	Hip-Hop	0.886***	(0.079)	0.012*	(0.005)	16926
	Rap	0.830***	(0.079)	0.011*	(0.005)	16464
	Indie Rock	0.776***	(0.074)	0.027	(0.017)	10878
Language	English	0.849***	(0.035)	0.006	(0.005)	76888
	Spanish	0.775***	(0.073)	0.018*	(0.009)	15036
	Portuguese	0.724***	(0.091)	0.001	(0.005)	12334
Artist Streams	1st Quartile	1.742***	(0.061)	0.003	(0.004)	42672
	2nd Quartile	1.047***	(0.044)	0.013**	(0.005)	42434
	3rd Quartile	0.763***	(0.041)	-0.001	(0.007)	42490
	4th Quartile	0.408***	(0.041)	0.018**	(0.006)	42504
Playlist Monthly Active Users	1st Quartile	0.652***	(0.047)	0.006	(0.006)	41496
	2nd Quartile	0.853***	(0.049)	0.004	(0.006)	41216
	3rd Quartile	0.995***	(0.051)	0.010	(0.005)	41356
	4th Quartile	1.407***	(0.058)	0.015**	(0.006)	41328

Notes: The estimates shown are for the coefficient of the interaction between *Playlisted* and *After Submission* as shown in Equation 1, and for two outcomes: All Streams (Log) and On Tour. The 1st quartile corresponds to the smallest number of artist streams and lowest number of playlist Monthly Active Users, respectively. The unit of observation is artist-month level. Clustered standard errors in parentheses at the artist level. All specifications have Artist and Month fixed effects.

Signif. Codes: *** $p < 0.001$, ** $p < 0.01$, * $p < 0.05$

It is plausible that playlisting may affect certain subgroups of artists more than others. We test for effects on subgroups defined by the artist's main genre, and the new release's reported language. Then, we test for differences in effects for artists with different levels of popularity, as defined by their pre-submission total streams. Finally, we test for differences based on the popularity of the playlist, as defined by the playlist's monthly active users. For each subgroup, we report the average effects on total streams and the likelihood of being on tour as specified in Equation 1, and include the corresponding event studies in the Appendix.

We specifically examine the genres that are most represented in our sample: Rap, Hip Hop, Indie-Rock and Pop. The results in Table 4 suggest that the effect of playlisting on streams appears to be significant and of varying magnitude for all genres listed. For the case of the likelihood of being on tour, we observe significant effects of around the same magnitude for most genres, with the exception of Indie Rock.

We examine the differential effects based on the reported language of the artist's submitted tracks. Specifically, we focus on English, Spanish, and Portuguese, the three predominant languages in our dataset. The average effects on streams and live performances are detailed in Table 4. All three languages show significant effects on streams. However, playlisted artists with Spanish tracks exhibit a statistically significant increase in the likelihood of touring, compared to those with English and Portuguese tracks, where the effect is positive but not statistically significant.

The multiplier effect of playlisting is hypothesized to be more pronounced for lesser-known artists who stand to gain more from enhanced discovery opportunities provided by playlists, as opposed to more established artists. Our study population is already restricted to emerging artists, those that have never signed with a major label. To further test this hypothesis, we divided our study population into quartiles based on their streaming counts in the month preceding a new release submission. The average streams for each quartile are presented in the Appendix, Table A5.

We estimate the effect of playlisting on streams for artists in each quartile, and report the average effects in Table 4, with the corresponding event study plots in the Appendix in Figure A9. The results suggest that artists with fewer streams before submission experience the most significant increase from playlisting. This effect tapers off as one progresses to quartiles with higher streaming counts, yet it remains statistically significant. We also estimate the effect for the likelihood of touring and note that artists in the highest quartile experience close to a 2 percentage point surge after their track gets playlisted.

We now investigate how treatment intensity, quantified by the monthly active users of the playlist, has an effect in the outcomes of the featured artists. For every artist included in a playlist, we recorded the monthly active users of that respective playlist one month prior to the new release submission. Subsequently, we group artists into quartiles according to the monthly active users of the playlist in which they were featured. In Table A6 we describe each quartile in terms of the average monthly active users for playlists in each group, as well as the average streams for artists featured on those playlists. We note that, while the differences in average stream counts across the quartiles are less pronounced than the differences shown in Table A5, the observed variation indicates a potential correlation between a playlist's monthly active users and the artist's streams. This suggests that editors might sort tracks into more prominent playlists based on an artist's prior success on the platform.

For each subgroup, we estimate the effect of playlisting in streams and the likelihood of touring, and report the average effects in Table 4, and the event study plots in the Appendix, Figures A11 and A12. The results reveal, perhaps unsurprisingly, that when an artist is added to a playlist

with a higher number of monthly active users, the increase in streams is higher. We do not find the same increasing pattern for the likelihood of going on tour, but we do note that artists in the highest quartile experience a slightly higher increase in their likelihood of touring.

7 Robustness Tests

7.1 Removal from Playlist

We have shown that after a new track is released and playlisted, artists experience an increase in streaming and spillovers into other on- and off-platform metrics compared to non-playlisted artists. To provide further evidence that this effect is driven by the playlist and not other unobserved factors, we examine the streams following the artist's removal from the playlist. We note that the timing of removal is endogenous and is often influenced by the song's performance or the playlist schedule. Therefore, we treat the evidence as suggestive and not a causal estimate.

In Figure 5a we show the monthly average streams for artists' music, broken down by the number of months the newly submitted track spent on the playlist, ranging from 0 (which corresponds to songs that were added and removed during the first month of release) and up to 6 months. We also show artists whose songs were not included in the playlist. We first note that regardless of how long the artists were featured, playlisted artists consistently exhibit higher streams on average compared to non-playlisted artists, a difference that persists months after their songs are removed from the playlists. The plot also shows a decline in overall streams after the song's removal from the playlist. We leverage the variability in the duration that songs spend on a playlist to formally measure the change in streams using a difference-in-difference event-study specification:

$$\begin{aligned} Streams_{it} = & \sum_{j=-6}^{-2} \delta_j \mathbf{1}(t - R_i = j) \\ & + \sum_{j=0}^{5} \delta_j \mathbf{1}(t - R_i = j) \\ & + \alpha_i + \lambda_t + \epsilon_{it} \end{aligned} \tag{4}$$

In this equation, $\mathbf{1}(t - R_i = j)$ is an indicator variable equal to one if month t is j months relative to the month of removal from the playlist for artist i. The coefficients δ_j capture the change in streams at each month j, measured relative to the omitted reference period, the month immediately before removal ($j = -1$). We include artist fixed effects α_i and month fixed effects λ_t, with time measured in months since the release of the new song.

Equation 4 presents the conventional two-way fixed effects event-study specification, but our estimates use the estimator proposed by Sun and Abraham (2021), which is robust to treatment effect heterogeneity across cohorts. The estimated coefficients are displayed in Figure 5b, confirming that there is a significant drop in streams that follows the song's removal from the playlist, that

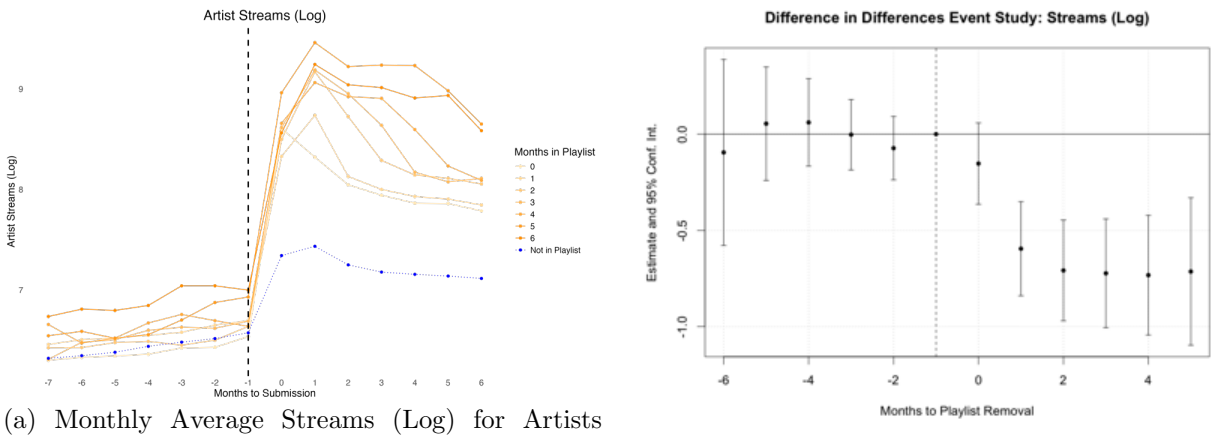
Table 5: Robustness Tests.

		(1)		(2)		(3)		
Panel A: On-platform Streams								
Category	Subset	All Streams (Log)		Other Music Streams (Log)		Old Music Streams (Log)		N
Full Sample	Full Sample	0.971***	(0.026)	0.389***	(0.022)	0.201***	(0.016)	170100
Post Covid	Post 2022-03	1.044***	(0.038)	0.408***	(0.032)	0.189***	(0.024)	81340
Time in playlist	7 days	0.760***	(0.047)	0.345***	(0.042)	0.165***	(0.031)	44240
	14 days	0.698***	(0.077)	0.317***	(0.07)	0.188***	(0.051)	14140
Time to playlist	Added at Release	0.932***	(0.035)	0.361***	(0.031)	0.196***	(0.023)	83048
	Month 1	1.017***	(0.039)	0.409***	(0.033)	0.201***	(0.024)	77840
	Month 2+	1.017***	(0.12)	0.544***	(0.111)	0.326***	(0.084)	7112
Panel B: Live-show Metrics								
Category	Subset	On Tour		Has Concert Page Views		Has "Find Tickets" Clicks		N
Full Sample	Full Sample	0.009**	(0.003)	0.010***	(0.003)	0.006	(0.003)	170100
Post Covid	Post 2022-03	0.008*	(0.004)	0.009*	(0.004)	0.005	(0.003)	81340
Time in playlist	7 days	0.001	(0.005)	0.004	(0.005)	-0.005	(0.005)	44240
	14 days	0.016	(0.009)	0.018	(0.01)	0.008	(0.011)	14140
Time to playlist	Added at Release	0.009*	(0.004)	0.011**	(0.004)	0.005	(0.004)	83048
	Month 1	0.006	(0.004)	0.006	(0.004)	0.004	(0.005)	77840
	Month 2+	0.035*	(0.016)	0.038*	(0.016)	0.038*	(0.016)	7112
Panel C: On-platform Engagement Metrics								
Category	Subset	Followers (Log)		Track Shares (Log)		User Playlist Adds (Log)		N
Full Sample	Full Sample	0.140***	(0.012)	0.557***	(0.018)	0.905***	(0.021)	170100
Post Covid	Post 2022-03	0.137***	(0.017)	0.568***	(0.024)	0.994***	(0.031)	81340
Time in playlist	7 days	0.120***	(0.022)	0.495***	(0.035)	0.744***	(0.04)	44240
	14 days	0.129***	(0.037)	0.493***	(0.057)	0.684***	(0.067)	14140
Time to playlist	Added at Release	0.125***	(0.017)	0.557***	(0.026)	0.903***	(0.03)	83048
	Month 1	0.150***	(0.017)	0.552***	(0.025)	0.909***	(0.032)	77840
	Month 2+	0.202***	(0.06)	0.659***	(0.094)	0.954***	(0.106)	7112

Notes: The estimates shown for the coefficient of the interaction between *Playlisted* and *After Submission* from Equation 1. The unit of observation is artist-month level. Clustered standard-errors in parentheses at the artist level. All specifications have Artist and Month fixed effects. The subset column indicates the subset of the data that is used to estimate the coefficients. Time in playlist subsamples estimates are for songs that spent exactly 7 and 14 days in the playlist.
Signif. Codes: *** $p < 0.001$, ** $p < 0.01$, * $p < 0.05$

starts in the month of removal.

Figure 5: Removal from Playlist Event Study.



(a) Monthly Average Streams (Log) for Artists Based on How Many Months They Were Included in the Playlist.

(b) Difference-in-Differences Event Study Plot Coefficients.

Note: The figure on the left shows the average number of monthly log streams for artists, broken down by the number of months the new song was featured in a playlist. Specifically, this plot corresponds to the streams of a subset of artists whose songs were released and included in any playlist in the month in which they were submitted to Spotify. The figure on the right shows the plot of the event study for the coefficients δ as described in Equation 4. The observed coefficients are relative to the month before the song was removed from the playlist.

7.2 Time in Playlist and Time to Playlist as Moderators

Our dataset is composed of playlists that completely refresh every week with a new selection of songs, and others that may carry the same set of tracks for months. In this section, we further leverage the variation in playlisting duration, but now we focus on how it moderates the effect that the playlist has on the artist's outcomes. We analyze this effect by estimating Equation 1 using two subsets of data, defined by the length of time the tracks were featured in the playlist. Specifically, we consider 7 days and 14 days, which represent the two most common time intervals in our sample. Estimates for all outcomes are shown in Table 5. As discussed in the previous section, given the non-random assignment of playlist durations, we cannot assess the causal impact of duration on outcomes. However, we observe that for streams and engagement metrics (Panels A and C), the results are significant, with only slight variations in magnitude between the two durations. For live show outcomes, we don't find significant estimates for those two durations.

The timing of when a track is featured on a playlist after its release could also influence the effect that the playlist has on the artist's outcomes. In other words, tracks added early after being released could see different effects on their streaming outcomes compared to those featured later. However, the timing of the addition to a playlist might not be random and can be influenced by factors such as the performance of the song, or the playlist refresh cycle. Acknowledging this limitation, we estimate Equation 1 using three subsets of the data, defined by the time when the

track was added to the playlist: added at the same time as released, added within the month of release, and added two months or more post-release. The results are presented in Table 5 for these three groups. The vast majority of tracks in our sample are added to playlists at release, alleviating concerns about track performance-driven selection bias. We find the effects to be of similar size across all groups, but particularly for tracks added at release and within the first month. Estimates for tracks added two or more months post-release are larger across all outcomes, though this group is smaller and exhibits larger standard errors.

7.3 COVID-19 Disruption

The COVID-19 pandemic significantly disrupted global concert availability due to government restrictions on live events and large gatherings, as shown in Appendix Figure A1. This disruption is particularly relevant to our analysis of live-show outcomes, as it may not reflect typical touring trends. It also relates to our streaming results, however, as the COVID-19 pandemic also led to a decline in global streaming activity during its early months (Sim et al., 2022a). Streaming changes early in the pandemic were attributed to changes in mobility for activities that complement music consumption, e.g. reduced commuting. In contrast, our data begins in March 2021, when streaming behaviors were expected to rebound closer to pre-pandemic norms. Google's COVID-19 Community Mobility Reports (Google, 2022) found that, by March 2021, mobility patterns had shifted closer to the pre-pandemic baseline levels.

To address concerns about changes in listener behavior during the pandemic, our main analysis sample is restricted to artists that submitted songs in late 2021, when some restrictions were relaxed and the COVID-19 vaccine was widely available. In addition, we also estimate the model only for a narrower set of artists who submitted tracks from March 2022 onward. By this time most restrictions had been lifted and both interest in and availability of live shows had increased. Estimates using this reduced set of artists are shown for all Panels in Table 5, under the "Post Covid" category. The results are similar in magnitude and significance to those estimated using the full sample.

7.4 Alternative Matching

In this study, we focused on a subset of 6,075 playlisted artists from our original sample of 7,738, identified through our matching process. Estimation on this subset might not correspond to the Average Treatment Effect on the Treated (ATT), primarily due to the exclusion of 1,663 treated units. This exclusion resulted from stringent exact matching criteria, including the track's submission month, genre, language, label type, and the artist's quantile group based on stream share. Particularly, the large number of genres and languages limited the number of potential matches for each treated artist. To address this, we implemented an alternative matching approach. Here, genre, language, label type, and artist quantile group were included in the propensity score calculation, but exact matching was applied only to the submission month. This method retained all 7,738 observations but occasionally paired treated and control artists differing in genre and

language. We show raw plots of the four outcomes (all streams, old streams, touring, followers) in appendix Figure A13 and all identify desirable pre-trends. Overall, results from the difference in differences specification using this alternative set of matched observations yields similar estimates in direction and size as in our reduced sample with more stringent matching. However we observed some differences in estimation, particularly for the estimates of live show likelihood across different language and genre subsets. For instance, while estimates for Rap and Hip-Hop remained consistent, the likelihood of touring for Pop artists decreased and lost statistical significance, whereas for Indie-Rock, it increased and became significant. Estimates for streaming and other on-platform outcomes remained stable and similar across both matching specifications.

8 Discussion and Conclusion

Our research explores how streaming platforms, via editorial decisions, can create positive multiplier effects on on- and off-platform activity of its content creators. We study the case of Spotify musicians who submitted tracks to Spotify's Playlist Pitching tool, a mechanism that allows musicians to put their new release in front of human editors for playlist consideration, along with relevant information of the musician and the track.

We find that artists that are added to these editorial playlists see an increase in their on- and off-platform activity, relative to non-playlisted artists. Specifically, playlisted artist's streams increase around 160%, and the likelihood of going on tour increases by around 0.9 percentage point, a 31% greater increase relative to non-playlisted artists. We also provide evidence of an increase in on-platform engagement, as measured by followers, unique listeners, shares and user-generated playlist additions. Contrary to beliefs about playlists failing to grow long-term engagement, we find evidence that editorial playlists can be a useful mechanism to create an engaged fan base, increasing both on-platform activity like shares, follows and user playlisting, and off-platform fan activity like live performances.

We also find that the multiplier effect from playlisting varies depending on the characteristics of the artists and playlisted songs, particularly popularity, genre and language. Even within our subset of artists, we find that those with the lowest stream counts pre-treatment seem to benefit the most from playlisting, in terms of on-platform activity. This is not the case for going on tour, where we find some weak evidence that artists in higher quartiles within our sample seem to benefit the most.

It is important to reiterate that the effects are estimated for creators that have been playlisted for the first time, are independent, and are at the long tail of the streaming distribution (i.e, they have a low number of streams before being playlisted) which might explain the dramatic increases observed. However, it is likely that the initial increase may allow the artist to gain momentum, and eventually lead the artist to be featured in other playlists, moving up in the playlist ecosystem. Furthermore, algorithmic playlists such as Release Radar and Discover Weekly may pick up on the increase in streams and recommend the artist (Bonini and Gandini, 2019) leading to more streams,

playlist features and fame.

Our study has several limitations. Even when we match on observable characteristics and identify a control group with very similar pre-trends to our playlisted group, we can not entirely rule out that the response to a new release will be larger for our playlisted group, regardless of playlist status. For example, if playlisted artists are consistently being promoted elsewhere at the same time as Spotify features them in an editorial playlists, these promotions could spillover into Spotify, increasing streams and biasing our results. We expect this to be less likely for the artists we study, emerging artists, who likely have a small (or no) promotional budget. Another concern is if editors are able to consistently pick "winners" (songs that would otherwise perform as well even if they were not playlisted). We attempt to control for extensive track level quality indicators, but we can not completely rule out a difference in quality.

While our paper has focused on editorial playlists, these are just one of the many ways that artists use streaming platforms to create an engaged fanbase and make their content more discoverable. Future research could assess the effectiveness of other mechanisms available to artists, such as using platform-based music tools and insights like Spotify for Artists on Spotify, to manage career decisions using fan demographics, as well as listener responses to releases, artist-created playlists, and new collaborations to support a career in music.

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9 Appendix

Figure A1: Normalized Number of Google Queries for the Term "Music Concerts".

Note: Query volume drops sharply from mid-March 2020 until early 2021. The period between the dotted red lines marks our study window. We only include queries made in the United States.

Table A1: Number of Playlists by Playlist Type.

Playlist Type	**Artists In Playlist**	**Percentage**
New Music & Emerging Artists	3,398	56%
Content Led	1,369	23%
Other	1,308	22%
Total	6,075	100%

9.1 Musical Covariates Used for Matching

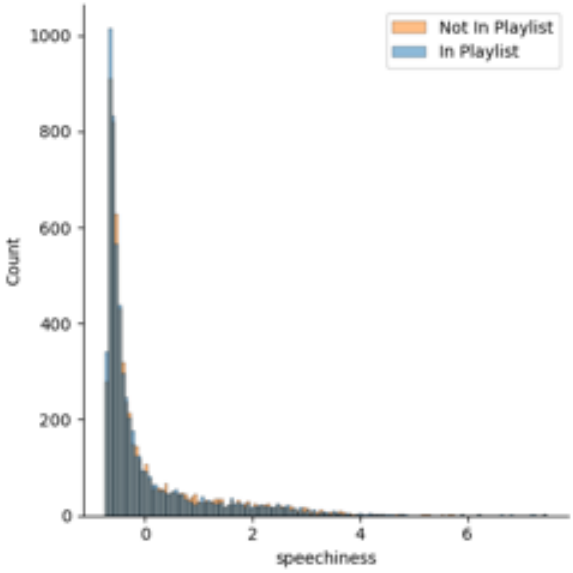
To find a control group that is similar to the playlisted tracks in our sample, we use propensity score matching. Table A2 shows a not exhaustive list of variables used for matching. We categorize the variables in three groups: lagged outcomes and other platform activity, self-reported track-level information that was submitted during the pitching process, and musical characteristics of the track.

The lagged outcomes, which include 6 months of streams, followers, monthly listeners, user-generated playlist additions, number of tracks, shares, the artist's rank, previous release activity, and concert page activity, will help us control for the artist's performance in the platform pre-submission. The self-reported track level information submitted to the Playlist Pitching tool will

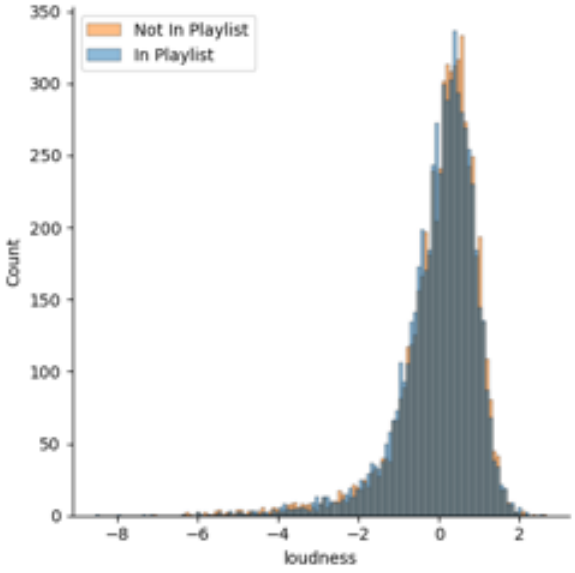
control for characteristics and context that are specific to the new tracks, such as the culture, theme, moods, languages, genre, and whether the track is a cover, instrumental, or a remix. Finally, the musical characteristics describe the song in 13 dimensions and serve as a proxy for the musical quality of the track. They include features describing the musical structure of the track, such as the mode (minor or major), the key, the tempo (speed of the track in beats per minute), time signature (which captures how many beats are in each bar), and the track duration; features describing the sonic texture and performance of the track, such as acousticness, instrumentalness (whether a track contains no vocals), energy (perceptual measure of intensity and activity), loudness (describing if the song is loud or soft in decibels), liveness (if the track is recorded in a studio or live), and speechiness (is it mostly instrumental or spoken); and features that capture the track's mood and physical or emotional response it might elicit, such as danceability (how suitable is the track for dancing, based on tempo, rhythm, beat etc.) and valance (is the track in the positive, happy, cheerful side or is it on the negative, sad, depressing, angry side).

Once we match, the resulting control group songs are almost indistinguishable in average musical features from the playlisted songs. In Table A4, we show the standardized mean differences, two-sided T-tests, and Mann Whitney U Test, and plot the histogram of the four musical features (speechiness, loudness, energy, and valance) with the largest standardized mean differences in Figure A2. The tests show that two features, speechiness and loudness, are statistically significant, but the difference is small, as evidenced by the standardized mean differences $< |0.05|$ and the histogram plots.

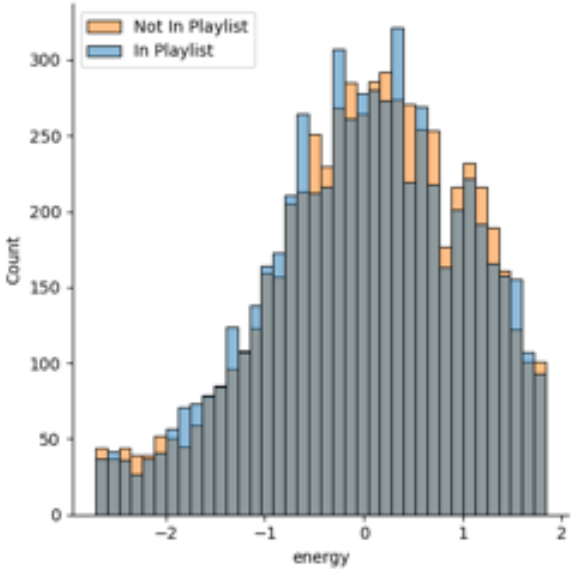
Figure A2: Histogram of Musical Features.



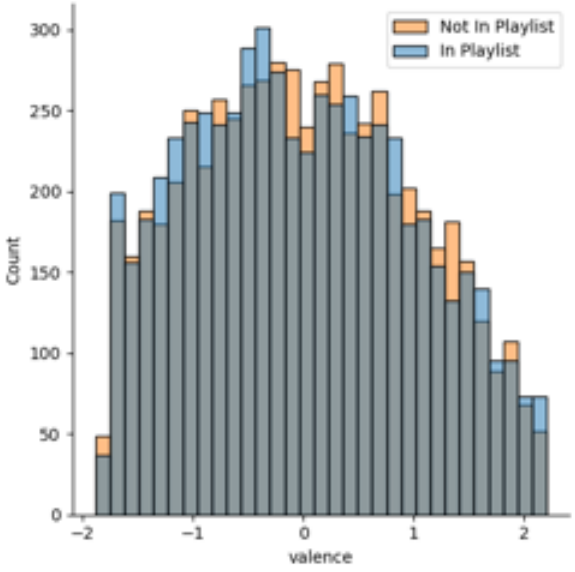
(a) Distribution of Normalized Speechiness.



(b) Distribution of Normalized Loudness.



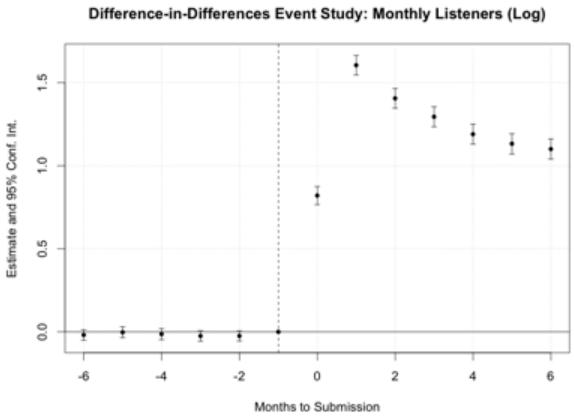
(c) Distribution of Normalized Energy.



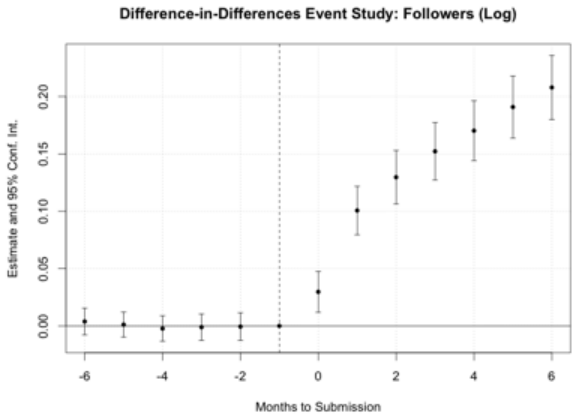
(d) Distribution of Normalized Valence.

Note: Histograms of the four musical features (speechiness, loudness, energy, valence) with the largest standardized mean differences, computed after performing propensity score matching. We show the histograms by playlist status. All musical feature scores have been normalized.

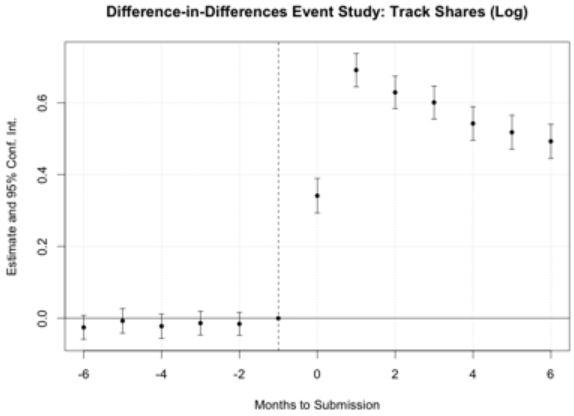
Figure A3: Event Study Plots for Artists' Engagement Metrics.



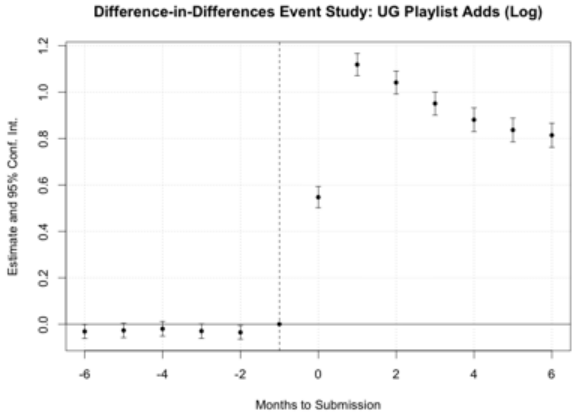
(a) Monthly Listeners (Log).



(b) Follower Count (Log).



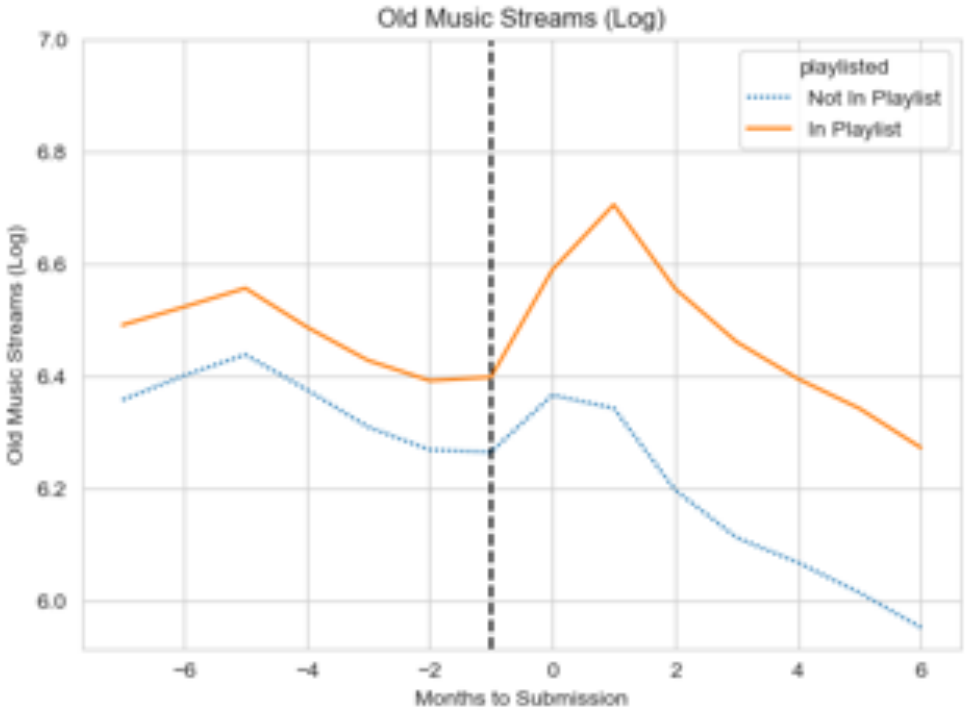
(c) Total Shares (Log).



(d) Total User-generated playlist Adds (Log).

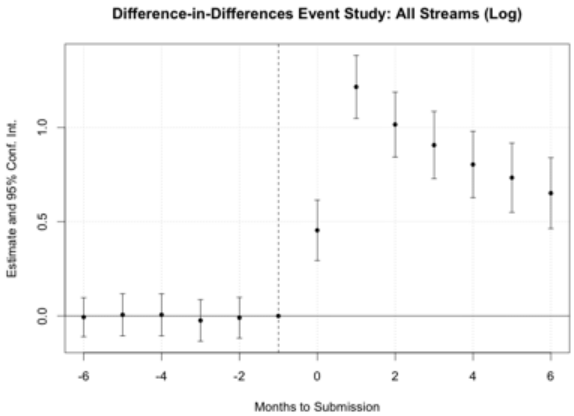
Note: The figure shows the event study estimates from Equation 2 where the dependent variable corresponds to one of the four engagement metrics: monthly listeners, follower count, total shares, and user generated playlist adds.

Figure A4: Average of Old Music Streams (Log).

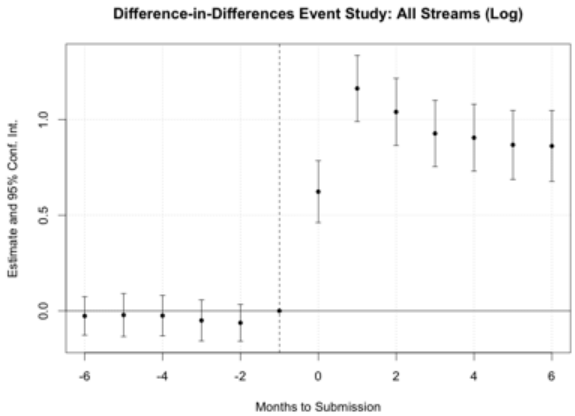


Note: The figure shows the average log streams for the artists' old music in the treatment and control groups.

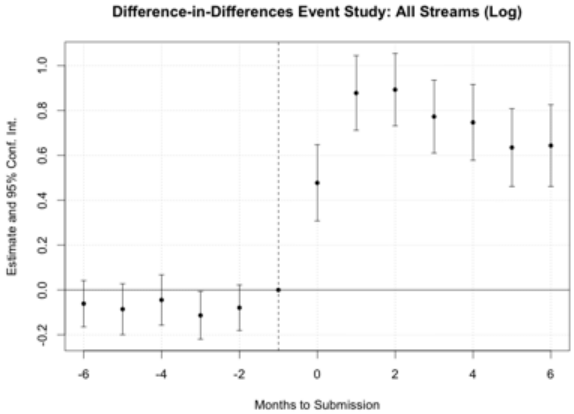
Figure A5: Event Study Plots for Artists' Streams by Genre.



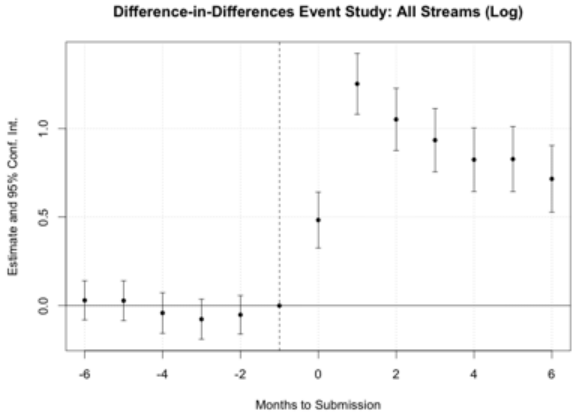
(a) Rap : All Streams (Log).



(b) Pop : All Streams (Log).



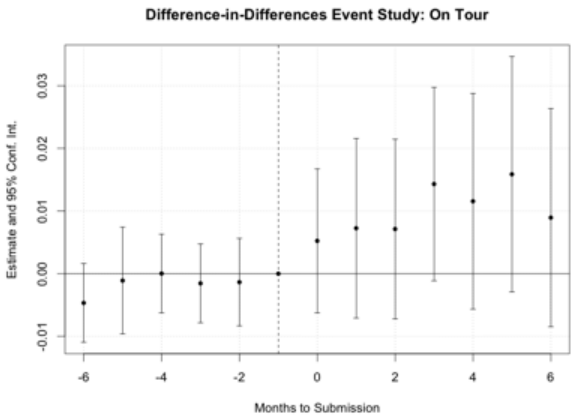
(c) Indie Rock : All Streams (Log).



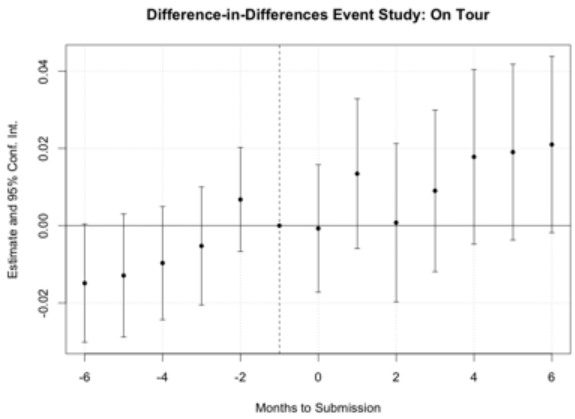
(d) Hip-Hop : All Streams (Log).

Note: The figure shows the event study estimates from Equation 2 where the dependent variable is the total number of log streams. Each figure corresponds to one of the top four genres in our sample.

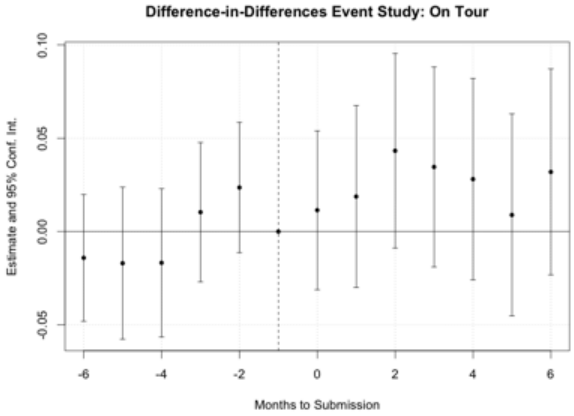
Figure A6: Event Study Plots for Artists' Probability of Being On Tour by Genre.



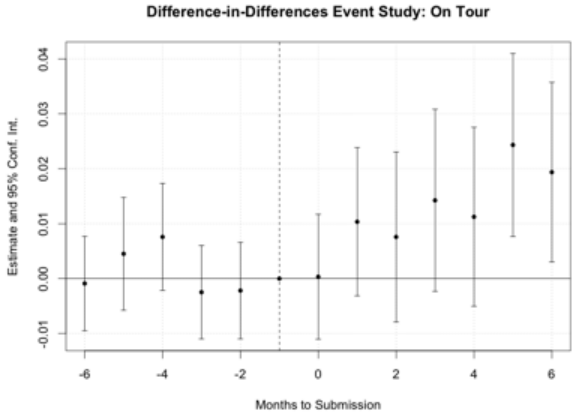
(a) Rap : Probability of Being On Tour.



(b) Pop : Probability of Being On Tour.



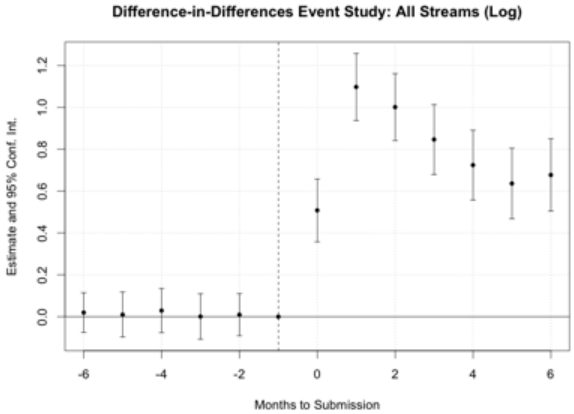
(c) Indie Rock : Probability of Being On Tour.



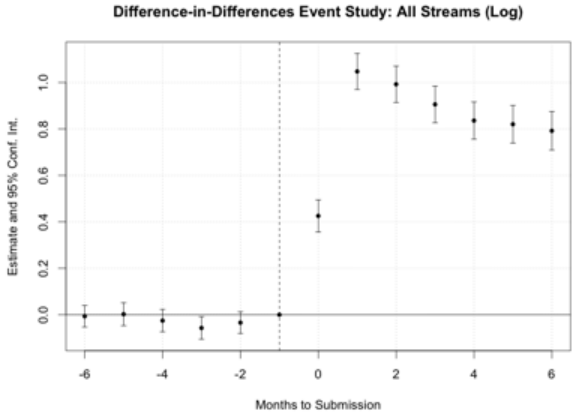
(d) Hip-Hop : Probability of Being On Tour.

Note: The figure shows the event study estimates from Equation 2 where the dependent variable is a binary indicator of whether the artist had any show scheduled in that month or any time in the future. Each figure corresponds to one of the top four genres in our sample.

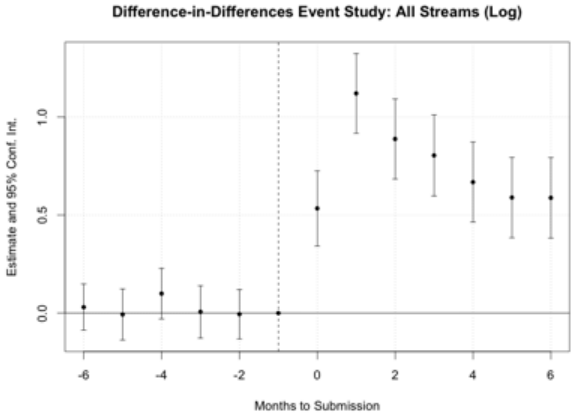
Figure A7: Event Study Plots for Artists' Streams by Track Language.



(a) Spanish: All Streams (Log).



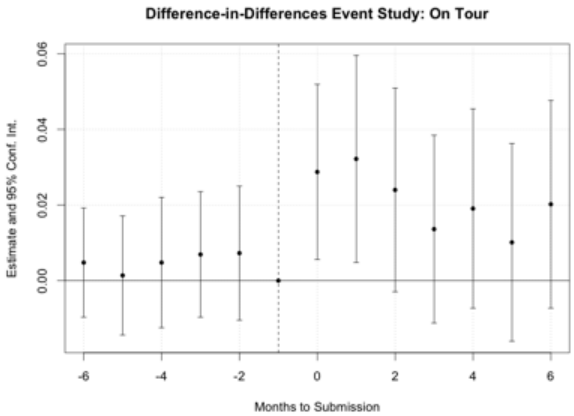
(b) English: All Streams (Log).



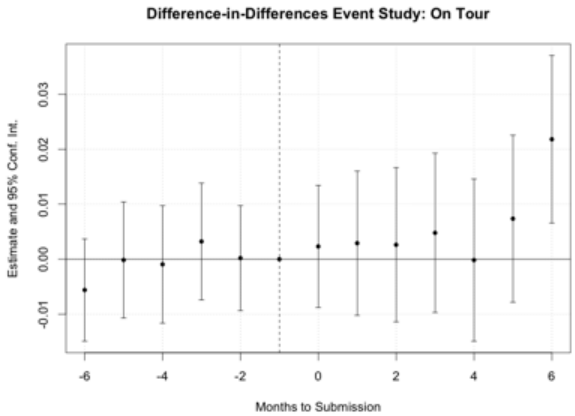
(c) Portuguese: All Streams (Log).

Note: The figure shows the event study estimates from Equation 2 where the dependent variable is the total number of log streams. Each figure corresponds to one of the top three track languages in our sample.

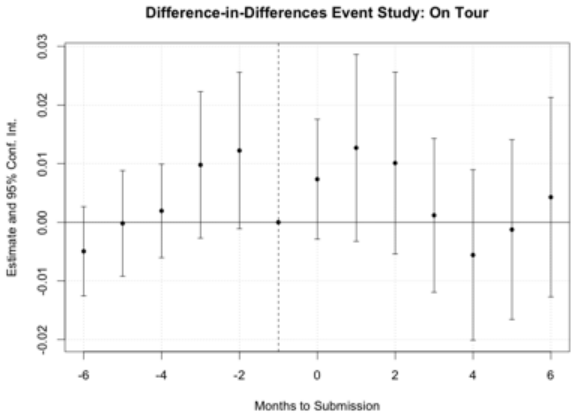
Figure A8: Event Study Plots for Artists' Probability of Being On Tour by Track Language.



(a) Spanish: Probability of Being On Tour.



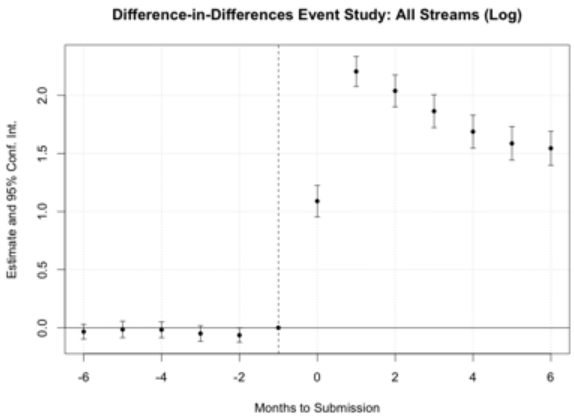
(b) English: Probability of Being On Tour.



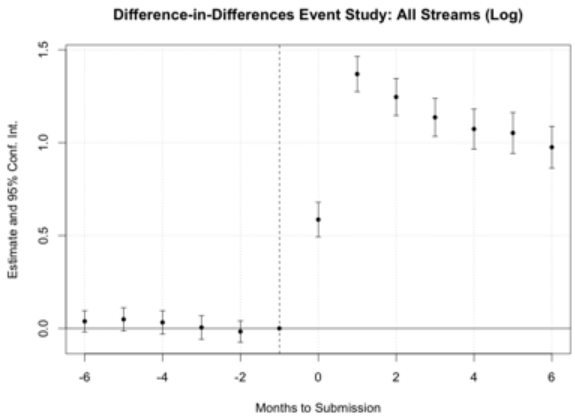
(c) Portuguese: Probability of Being On Tour.

Note: The figure shows the event study estimates from Equation 2 where the dependent variable is a binary indicator of whether the artist had any show scheduled in that month or any time in the future. Each figure corresponds to one of the top three track languages in our sample.

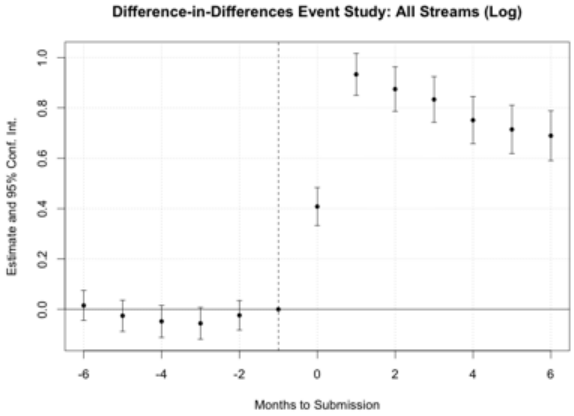
Figure A9: Event Study Plots for Artists' Streams by Artist Quartiles.



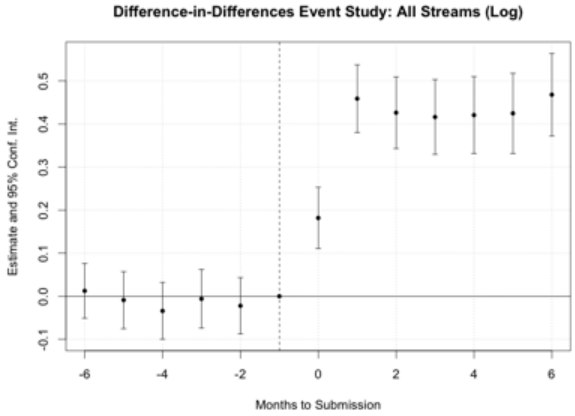
(a) Artist Quartile 1: All Streams (Log).



(b) Artist Quartile 2: All Streams (Log).



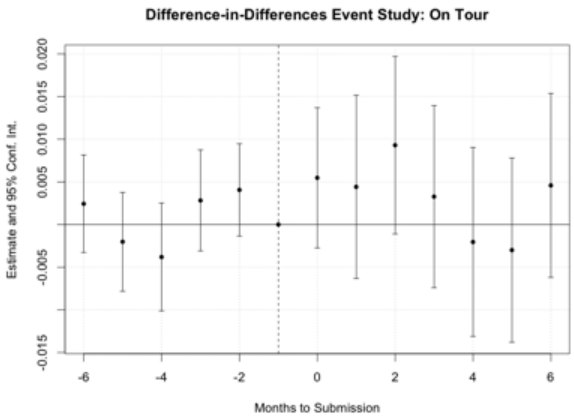
(c) Artist Quartile 3: All Streams (Log).



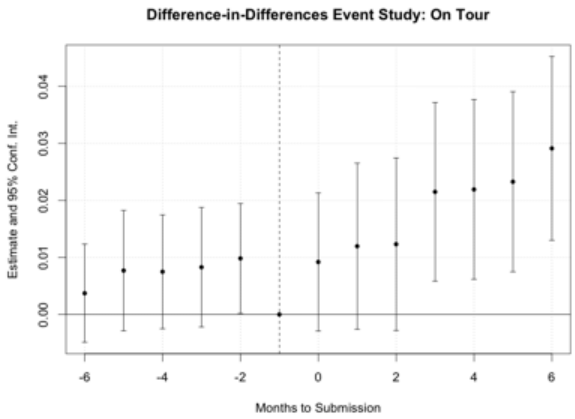
(d) Artist Quartile 4: All Streams (Log).

Note: The figure shows the event study estimates from Equation 2 where the dependent variable is the total number of log streams. Each figure corresponds to one of the artist quartile groups, based on the artist's streams before submission.

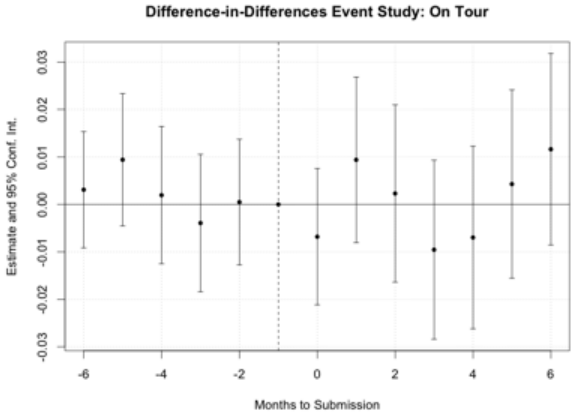
Figure A10: Event Study Plots for Artists' Probability of Being On Tour by Artist Quartiles.



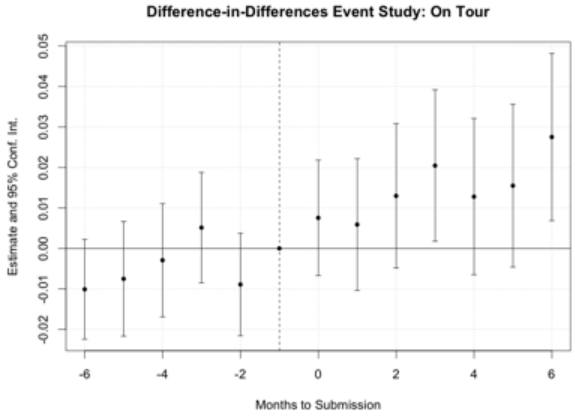
(a) Artist Quartile 1: Probability of Being On Tour.



(b) Artist Quartile 2: Probability of Being On Tour.



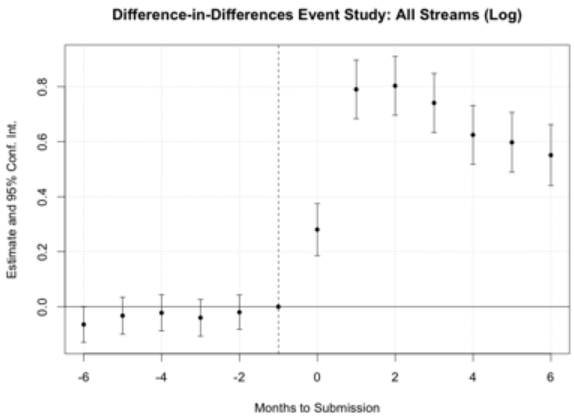
(c) Artist Quartile 3: Probability of Being On Tour.



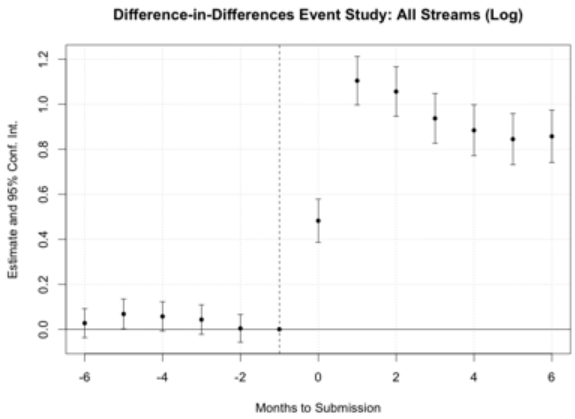
(d) Artist Quartile 4: Probability of Being On Tour.

Note: The figure shows the event study estimates from Equation 2 where the dependent variable is a binary indicator of whether the artist had any show scheduled in that month or any time in the future. Each figure corresponds to one of the artist quartile groups, based on the artist's streams before submission.

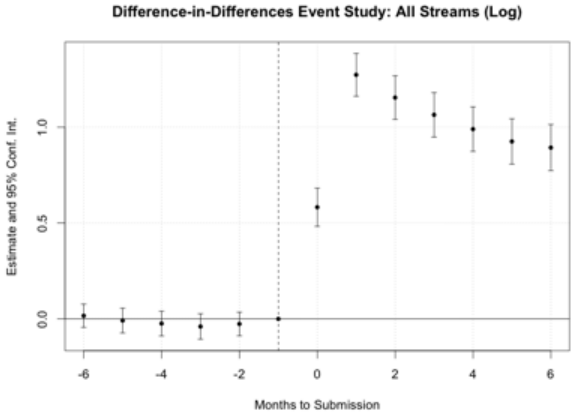
Figure A11: Event Study Plots for Artists' Streams by Playlist Quartiles.



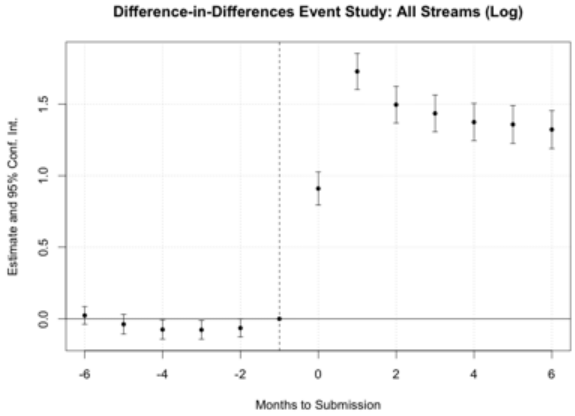
(a) Playlist Quartile 1: All Streams (Log).



(b) Playlist Quartile 2: All Streams (Log).



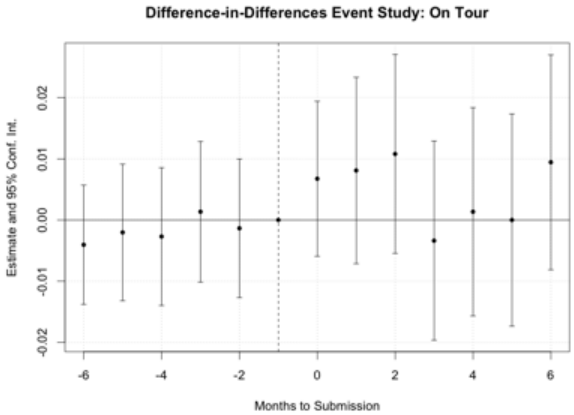
(c) Playlist Quartile 3: All Streams (Log).



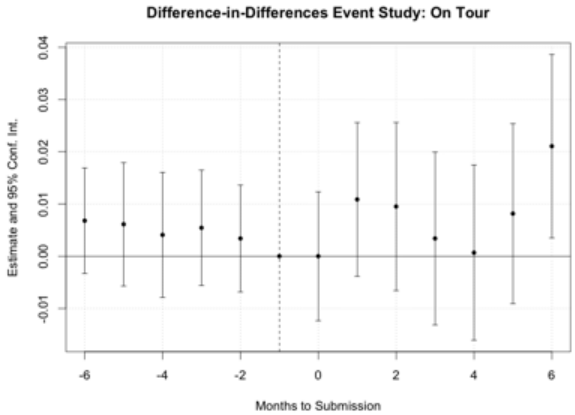
(d) Playlist Quartile 4: All Streams (Log).

Note: The figure shows the event study estimates from Equation 2 where the dependent variable is the total number of log streams. Each figure corresponds to one of the playlist quartile groups, based on the playlist's monthly active users in the month before submission.

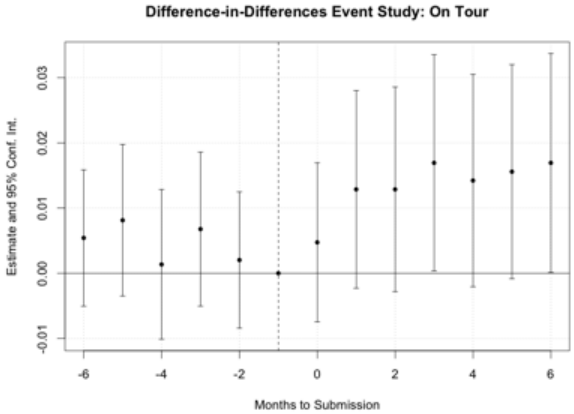
Figure A12: Event Study Plots for Artists' Probability of Being On Tour by Playlist Quartiles.



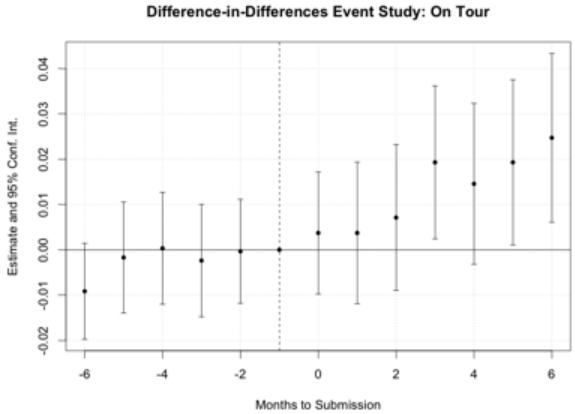
(a) Playlist Quartile 1: Probability of Being On Tour.



(b) Playlist Quartile 2: Probability of Being On Tour.



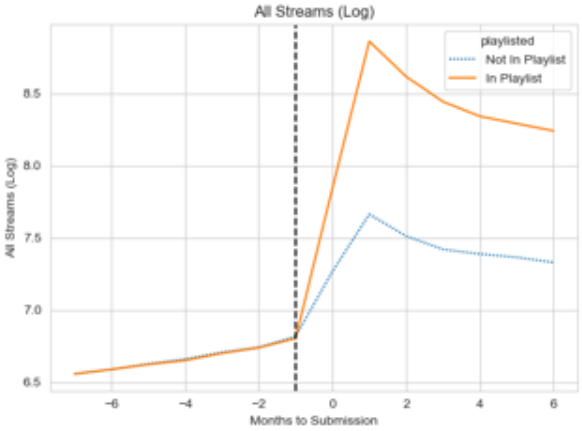
(c) Playlist Quartile 3: Probability of Being On Tour.



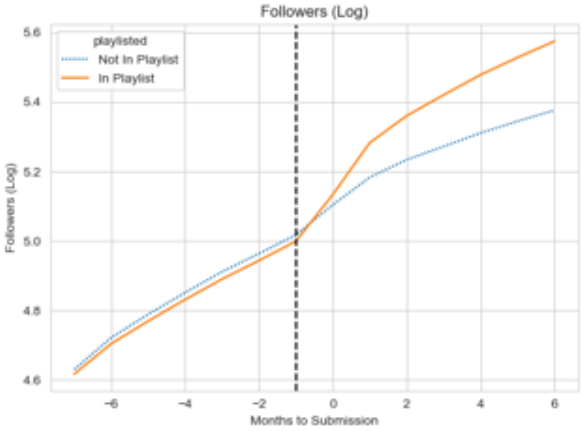
(d) Playlist Quartile 4: Probability of Being On Tour.

Note: The figure shows the event study estimates from Equation 2 where the dependent variable is a binary indicator of whether the artist had any show scheduled in that month or any time in the future. Each figure corresponds to one of the playlist quartile groups, based on the playlist's monthly active users in the month before submission.

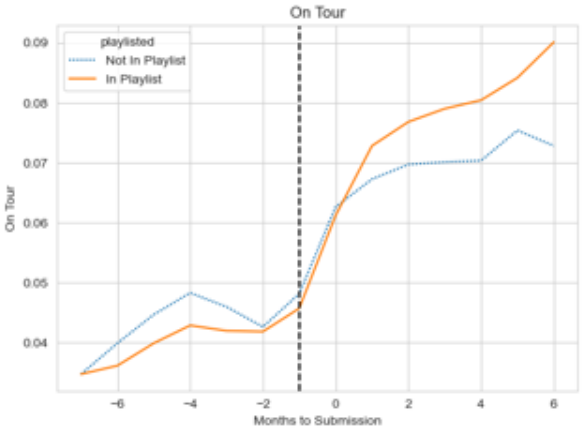
Figure A13: Raw Data Plots for Four Outcomes of Interest, Using an Alternative Matching Method.



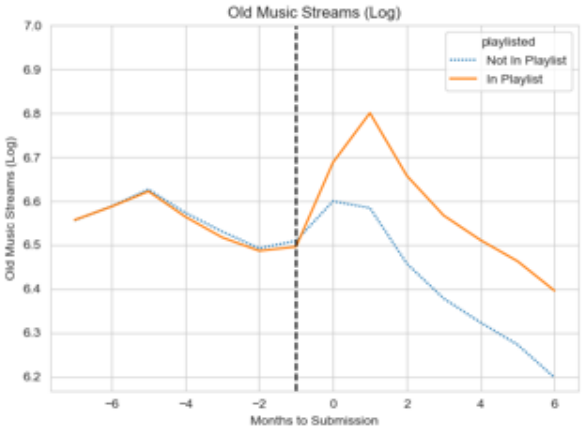
(a) All Streams (Log).



(b) Follower Count (Log).



(c) Touring.



(d) Old Music Streams (Log).

Table A2: List of Variables Used in Propensity Score Matching (Not Exhaustive).

Artist and Track Level Covariates used for Propensity Score Matching
Lagged Outcomes and Platform Activity
All Streams (Log)*
Old Streams 30s Split (Log)*
Tracks With Streams*
Followers (Log)*
Monthly Listeners (Log)*
On Tour (Binary) *
UG Playlists Adds (Log)*
Total Shares (Log)*
Total Release Radar Adds*
Has Concert Page Views (Binary)*
Has "Find Tickets" Clicks (Binary)*
Concert-Page Views (Log)*
"Find Tickets" Clicks (Log)*
Last 90 day Streams (Log)
Artist Rank
Days Since Last Release
Years Since First Release
Most Recent Licensor Category
Artist Gender (self reported)
Self-reported track-level information when pitching track
Cultures
Themes
Moods
Languages
Track's main Genre
Is Track a Cover
Is Track Instrumental
Is Track a Remix
Recording Type
Submitter Organization
Number of Instruments
Free Text Description Word Count
Musical characteristics of track submitted to playlist pitching mechanism
Danceability
Energy
Key
Loudness
Mode
Speechiness
Acousticness
Instrumentalness
Liveness
Valence
Tempo
Duration (ms)
Time Signature

Note: Contact the authors for the full list of matching variables.
Note: Variables marked with an asterisk (*) include measurements over the six periods prior to the new song submission.

Table A3: Summary Statistics: Means, Standardized Mean Difference.

Variable Name	Mean Playlisted	Mean Non-Playlisted (Un-matched)	Std. Mean Diff.	Mean Non-Playlisted (Matched)	Std. Mean Diff.
Artist Level Metrics					
Distance	0.1158	0.0313	0.6614	0.0705	0.2026
All Streams (Month=-1)	6.8063	5.8136	0.4232	6.6033	0.0456
Old Streams (Month=-1)	6.4990	5.4374	0.4539	6.2660	0.0570
Tracks With Streams (Month=-1)	17.9475	17.4821	0.0127	17.6545	-0.0183
Followers Log (Month=-1)	5.0016	4.4217	0.3372	4.8805	0.0424
Monthly Listeners Log (Month=-1)	5.8769	4.6937	0.5017	5.6460	0.0654
On Tour (Month=-1)	0.0459	0.0316	0.0683	0.0423	0.0118
User Playlists Adds (Log) (Month=-1)	3.3326	2.2912	0.4940	3.0845	0.0846
Total Shares (Log) (Month=-1)	1.8119	1.2424	0.3668	1.6322	0.0707
Has Concert Page Views (Month=-1)	0.0510	0.0337	0.0786	0.0479	0.0112
Has "Find Tickets" Clicks (Month=-1)	0.0423	0.0217	0.1024	0.0375	0.0270

Notes: Only a subset of the relevant covariates are presented in this table.

Table A4: Standardized Mean Differences and p-Values From Two-Sample t-Tests and Mann–Whitney U Tests for Differences in Musical Features.

Feature	SMD	T-test p-value	Mann–Whitney U p-value
speechiness	0.0400	0.0274	0.0004
loudness	0.0313	0.0842	0.0010
energy	0.0290	0.1096	0.0549
valence	0.0204	0.2618	0.2003
time signature	0.0112	0.5359	0.3455
instrumentalness	0.0105	0.5634	0.7987
mode	0.0096	0.5962	0.5962
danceability	0.0033	0.8566	0.7085
duration in milliseconds	0.0021	0.9084	0.4112
liveness	0.0016	0.9286	0.5874
tempo	0.0015	0.9350	0.6960
key	-0.0050	0.7841	0.8134
acousticness	-0.0117	0.5186	0.7566

Table A5: Average Streams for Artists in Each Streaming Quartile.

Streaming Quartile	Artists	All Streams	All Streams (Log)
1st Quartile	3,048	57.48 (40.57)	3.72 (0.95)
2nd Quartile	3,031	371.23 (163.57)	5.82 (0.46)
3rd Quartile	3,035	1,896.72 (939.08)	7.43 (0.5)
4th Quartile	3,036	23,748.82 (22911.48)	9.67 (0.9)

Note: Quartile is assigned based on the artist's number of streams one month before submission

Table A6: Average Streams for Artists in Each Quartile, as Defined by the Playlist's Monthly Active Users.

Artist-Playlist Quartile	Artists	Playlist Monthly Active Users	All Streams	All Streams (Log)
1st Quartile	1,482	7,007.75 (3706.79)	3,998.06 (11890.94)	6.13 (2.15)
2nd Quartile	1,472	22,823.75 (6068.5)	5,938.8 (13953.84)	6.56 (2.3)
3rd Quartile	1,477	57,459.6 (15852.2)	8,273.4 (17777.4)	6.97 (2.33)
4th Quartile	1,476	257,843.67 (349267.63)	9,558.91 (19492.65)	7.16 (2.33)

Table A7: OLS Regression Results Using the New Track Release as the Post-Treatment Indicator.

Panel A: On-platform Streams			
Dependent Variable	All Streams (Log)	Other Music Streams (Log)	Old Music Streams (Log)
Playlisted x After Released	0.932***	0.364***	0.197***
	(0.025)	(0.021)	(0.016)
Panel B: Live Show Metrics			
Dependent Variable	On Tour	Has Concert Page Views	Has "Find Tickets" Clicks
Playlisted x After Released	0.006*	0.007*	0.002
	(0.003)	(0.003)	(0.003)
Panel C: Engagement Metrics			
Dependent Variable	Followers (Log)	Track Shares (Log)	UG Playlist Adds (Log)
Playlisted x After Released	0.124***	0.544***	0.869***
	(0.011)	(0.017)	(0.02)
Observations (N): 149,942			

Notes: The table reports OLS estimates of the interaction between *Playlisted* and *After Released*, similar to Equation 1, but using the new track release as the post-treatment indicator. Results are similar in magnitude and significance, as reported in Table 3. The unit of observation is at the artist-month level. Standard errors clustered at the artist level are in parentheses. All regressions include artist and month fixed effects.
Signif. Codes: *** $p < 0.001$, ** $p < 0.01$, * $p < 0.05$